

Epigenetic Control of Temperature Response in *Ostreococcus tauri* (ECTOR)

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H3K27me3 genome wide distribution analysis

We load the genome of *Ostreococcus tauri* and compute the length of its chromosomes. We determine the length of the largest chromosome and for technical reasons we reverse the vector of lengths. This will be used to represent graphically the genome wide distribution of H3K27me3.

```
library(seqinr)

ostta.genome <- read.fasta(file = "ostta_genome/ostreococcus_tauri.fa", seqtype = "DNA")
chr.lens <- sapply(X = getSequence(ostta.genome), FUN = length)

number.chrs <- length(getSequence(ostta.genome))
max.chr.len <- max(chr.lens)
chr.lens <- rev(chr.lens)
```

We load the H3K27me3 peaks identified in cultures grown under constant light at 20°C. We represent each chromosome using a rectangle. Each H3K27me3 peak is represented in its precise genome location using a blue rectangle.

```
ll.20 <- read.table(file="chipseq_data/bed_data/h3k27me3/h3k27me3_ll_20.bed",
                    header = F)

ll.20 <- ll.20[,1:3]

nrow(ll.20)

## [1] 480

#jpeg(filename = "images/h3k27me3_genome_wide_distribution_ll_20.jpg", res = 300, width = 900, height = 900)
#par(mar=c(0,0,0,0))
plot(x=c(0,max.chr.len), y=c(0,4*number.chrs), col = "white", xlab = "", ylab = "", axes=F)

for(i in 1:number.chrs)
{
  polygon(x = c(0, chr.lens[i], chr.lens[i], 0),
          y = c(4*i+1, 4*i+1, 4*i-1, 4*i-1),
          col = "white")
}

for(i in 1:nrow(ll.20))
{
  current.chr <- ll.20[i,1]
```

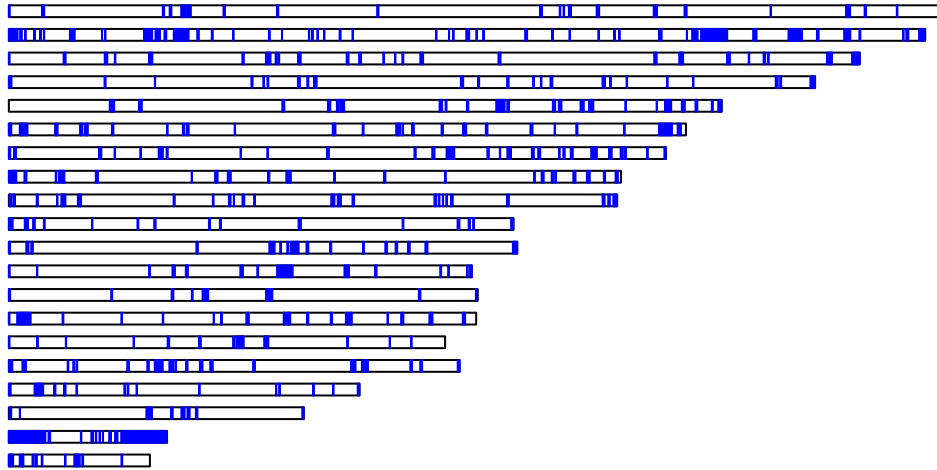
```

current.start <- ll.20[i,2]
current.end <- ll.20[i,3]

current.line <- (20:1)[current.chr]

polygon(x = c(current.start, current.end, current.end, current.start),
        y = c(4*current.line+1, 4*current.line+1, 4*current.line-1, 4*current.line-1),
        col = "blue",border="blue")
}

```



```
#dev.off()
```

We identified 480 genomic regions significantly occupied by H3K27me3 in the three different independent biological replicates. We compute the percentage of the whole genome occupied by H3K27me3 and the percentage in each chromosome. A barplot is used to represent the percentage of H3K27me3 in each chromosome. Z-scores and p-values are computed for each chromosome. A gradient from white to blue is used to represent significance levels.

```

h3k27me3.peaks.len <- ll.20$V3 - ll.20$V2
sum(h3k27me3.peaks.len)/sum(chr.lens)

```

```
## [1] 0.07163182
```

```

ll.20 <- cbind(ll.20,h3k27me3.peaks.len)
colnames(ll.20) <- c("chr","start","end","length")
head(ll.20)

```

```

##   chr  start    end length
## 1   1    174   1061    887
## 2   1 38825 41425   2600
## 3   1 180446 180824    378
## 4   1 181798 181979    181
## 5   1 188479 190520   2041
## 6   1 202039 207172   5133

```

```

number.of.h3k27.peaks.chr <- vector(mode = "numeric",length = 20)
peaks.len.chr <- vector(mode = "numeric",length = 20)

```

```

for(i in 1:20)
{
  number.of.h3k27.peaks.chr[i] <- nrow(subset(ll.20, chr == i))
}

```

```

peaks.len.chr[i] <- sum(subset(ll_20, chr == i)[,4])
}

percentage.chr.k27 <- peaks.len.chr/rev(chr.lens)

z.scores <- (percentage.chr.k27 - mean(percentage.chr.k27))/sd(percentage.chr.k27)

which(z.scores > 2.5)

## [1] 19

p.vals <- pnorm((percentage.chr.k27 - mean(percentage.chr.k27[-19]))/sd(percentage.chr.k27[-19]),lower.tail=FALSE)

p.vals

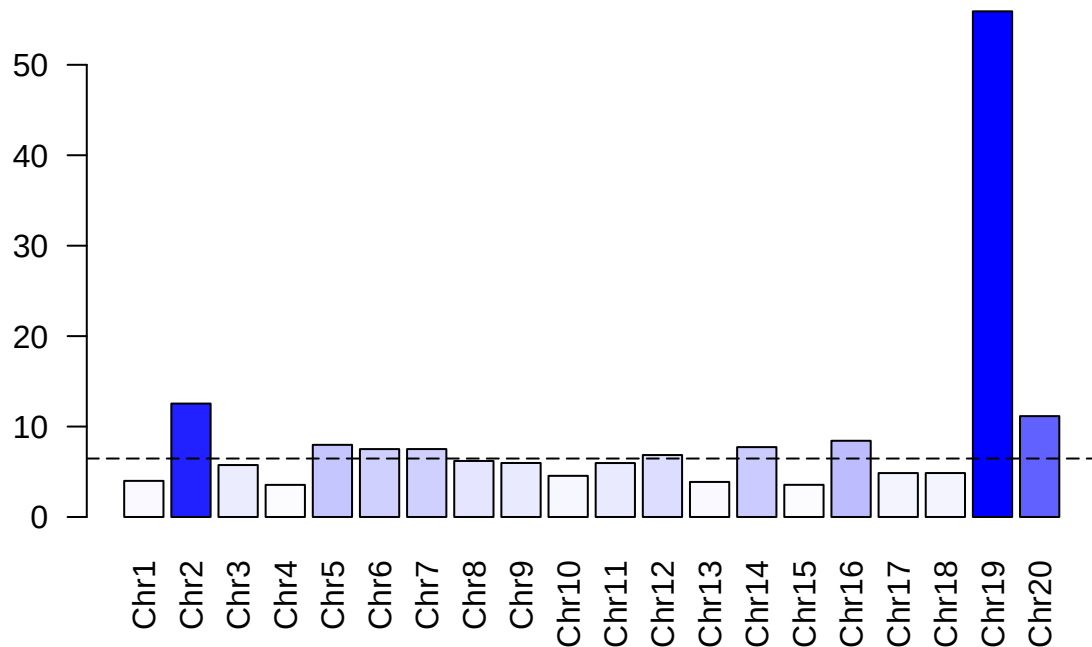
## [1] 8.428706e-01 6.683267e-03 6.150346e-01 8.821057e-01 2.690296e-01
## [6] 3.356068e-01 3.348786e-01 5.436769e-01 5.792470e-01 7.820867e-01
## [11] 5.803782e-01 4.374461e-01 8.540130e-01 3.039144e-01 8.815987e-01
## [16] 2.119113e-01 7.434354e-01 7.430300e-01 2.062177e-90 2.815232e-02

values.for.colors <- round(-log10(p.vals)*40)
values.for.colors[19] <- 100

white.blue.palette <- colorRampPalette(c('white','blue'))

#jpeg(filename = "images/h3k27me3_chr_barplot_ll_20.jpg",res = 300,width = 1800,height = 900)
barplot(100*percentage.chr.k27,border="black",names.arg = paste0("Chr",1:20),las=2,col=white.blue.palette[values.for.colors])
abline(h = 100*mean(percentage.chr.k27[-19]),lty=5)

```



```
#dev.off()
```

Approximately 7.2% of the *Ostreococcus* genome is occupied by H3K27me3. H3K27me3 is mostly uniformly distributed over the different chromosome although it significantly accumulates (p-value < 0.01) in Chr2 (sexual chromosome) and Chr19 (viral resistance chromosome) with p-values 6.68×10^{-3} and 2.06×10^{-90} respectively.

We determine the gene features occupied by each H3K27me3 mark and the H3K27me3 marked genes.

```
library(TxDb.Otauri.JGI)
```

```
## Loading required package: GenomicFeatures
## Loading required package: BiocGenerics
##
## Attaching package: 'BiocGenerics'
## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs
## The following objects are masked from 'package:base':
##
##     anyDuplicated, append, as.data.frame, basename, cbind, colnames,
##     dirname, do.call, duplicated, eval, evalq, Filter, Find, get, grep,
##     grepl, intersect, is.unsorted, lapply, Map, mapply, match, mget,
##     order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##     rbind, Reduce, rownames, sapply, setdiff, sort, table, tapply,
##     union, unique, unsplit, which.max, which.min
## Loading required package: S4Vectors
## Loading required package: stats4
##
## Attaching package: 'S4Vectors'
## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname
## Loading required package: IRanges
## Loading required package: GenomeInfoDb
## Loading required package: GenomicRanges
## Loading required package: AnnotationDbi
## Loading required package: Biobase
## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname)".
txdb <- TxDb.Otauri.JGI

ostta.genes <- as.data.frame(genes(txdb))
head(ostta.genes)
```

```
##           seqnames start   end width strand   gene_id
## ostta01g00010      1    244  1932  1689      - ostta01g00010
## ostta01g00020      1   2077   4029  1953      + ostta01g00020
## ostta01g00030      1   4157   5299  1143      + ostta01g00030
## ostta01g00040      1   5392   7041  1650      - ostta01g00040
## ostta01g00050      1   7386   9605  2220      - ostta01g00050
```

```
## ostta01g00060      1  9780 10412   633      - ostta01g00060
target.genes <- vector(mode = "character",length = nrow(ll.20))
peak.annotation <- vector(mode = "character",length = nrow(ll.20))

for(i in 1:nrow(ll.20))
{
  current.chr <- ll.20[i,1]
  current.start <- ll.20[i,2]
  current.end <- ll.20[i,3]

  complete.gene.body <- subset(ostta.genes,
                              seqnames == current.chr &
                              start >= current.start &
                              end <= current.end)

  overlap.tss.pos.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   start >= current.start & start <= current.end &
                                   strand == "+")

  overlap.tss.neg.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   end >= current.start & end <= current.end &
                                   strand == "-")

  overlap.tes.pos.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   end >= current.start & end <= current.end &
                                   strand == "+")

  overlap.tes.neg.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   start >= current.start & start <= current.end &
                                   strand == "-")

  inside.gene.body <- subset(ostta.genes, seqnames == current.chr &
                             start <= current.start & end >= current.end)

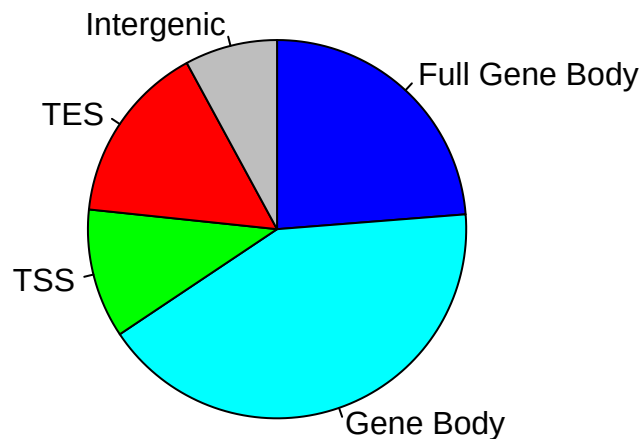
  if(nrow(complete.gene.body) > 0)
  {
    target.genes[i] <- paste(unlist(complete.gene.body$gene_id),collapse = ",")
    peak.annotation[i] <- "gene.body"
  } else if(nrow(overlap.tss.pos.strand) > 0)
  {
    target.genes[i] <- paste(unlist(overlap.tss.pos.strand$gene_id),collapse = ",")
    peak.annotation[i] <- "TSS"
  } else if(nrow(overlap.tss.neg.strand) > 0)
  {
    target.genes[i] <- paste(unlist(overlap.tss.neg.strand$gene_id),collapse = ",")
    peak.annotation[i] <- "TSS"
  } else if(nrow(overlap.tes.pos.strand) > 0)
  {
```

```

target.genes[i] <- paste(unlist(overlap.tes.pos.strand$gene_id),collapse = ",")
peak.annotation[i] <- "TES"
} else if(nrow(overlap.tes.neg.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tes.neg.strand$gene_id),collapse = ",")
  peak.annotation[i] <- "TES"
} else if(nrow(inside.gene.body) > 0)
{
  target.genes[i] <- paste(unlist(inside.gene.body$gene_id),collapse = ",")
  peak.annotation[i] <- "internal.gene.body"
} else
{
  peak.annotation[i] <- "intergenic"
}
}

freq.peak.annotation <- table(peak.annotation)
#jpeg(filename = "images/h3k27me3_pie_chart_ll_20.jpg",res = 300,width = 900,height = 900)
#par(mar=c(0,0,0,0))
pie(freq.peak.annotation[c("gene.body",
                           "internal.gene.body",
                           "TSS",
                           "TES",
                           "intergenic")],
    labels = c("Full Gene Body",
               "Gene Body",
               "TSS",
               "TES",
               "Intergenic"),
    clockwise = T, border = "black", col=c("blue","cyan","green","red","grey"))

```



```

#dev.off()

percentage.freq.peak.annotation <- 100*freq.peak.annotation/sum(freq.peak.annotation)

percentage.freq.peak.annotation["gene.body"] +
percentage.freq.peak.annotation["internal.gene.body"]

## gene.body

```

```
##      65.625
percentage.freq.peak.annotation

## peak.annotation
##      gene.body      intergenic internal.gene.body      TES
##      23.750000      7.916667      41.875000      15.416667
##      TSS
##      11.041667

ll.20.peak.annotation <- cbind(ll.20[,1:3],data.frame(peak.annotation,target.genes))

head(ll.20.peak.annotation)

##   chr  start    end peak.annotation target.genes
## 1   1    174   1061          TES ostta01g00010
## 2   1   3825  41425      gene.body ostta01g00260
## 3   1 180446 180824      intergenic
## 4   1 181798 181979      intergenic
## 5   1 188479 190520 internal.gene.body ostta01g01270
## 6   1 202039 207172          TSS ostta01g01340

write.table(x = ll.20.peak.annotation,file = "tables/H3K27me3_LL_20C_peak_annotation.tsv",quote = F,sep

all.k27.genes <- (unlist(sapply(X =ll.20.peak.annotation$target.genes,
                               FUN = function(x) {strsplit(x,",")[[1]]})))

names(all.k27.genes) <- NULL
length(all.k27.genes)

## [1] 491

length(all.k27.genes)/7668

## [1] 0.06403234
```

We found 491 genes associated to H3K27me3 marks corresponding to 6.4% of the entire *Ostreococcus* genome. H3K27me3 is mostly located occupying either the entire gene body (23.75%) or internal regions of gene body (41.86%). Although some regions occupied by H3K27me3 has been found overlapping only with the TSS (11.04%) or the TES (15.42%). H3K27me3 was found in intergenic regions only in 7.92% of the cases.

Next we use metageneplots to determine the shape of the H3K27me3 over its occupied genes. We consider three different sets of marked genes, genes with their entire gene body marked, genes with H3K27 inside their gene body with no overlap in the TSS or TES and genes with H3K27me3 overlapping their TSS or TES

```
## Function to compute ChIP signal over different tiles
ntile.signal <- function(signal.vector,number.tiles)
{
  step.len <- floor(length(signal.vector)/number.tiles)
  number.plus.1 <- length(signal.vector) %% number.tiles
  pos.plus.1 <- sort(sample(1:100,number.plus.1,replace = F))

  acum <- c(0)
  for(i in 1:number.tiles)
  {
    if(i %in% pos.plus.1)
    {
      acum <- c(acum,acum[i]+step.len+1)
    } else
  }
```

```

{
  acum <- c(acum,acum[i]+step.len)
}
}

acum[1] <- 1

n.tile.signal.vector <- vector(mode = "numeric",length = number.tiles)
for(i in 2:length(acum))
{
  n.tile.signal.vector[(i-1)] <- mean(signal.vector[acum[i-1]:acum[i]])
}

return(n.tile.signal.vector)
}

## Function to generate the data for a metageneplot
metageneplot <- function(gene.names,genes,bw.file.name,ntile.txs,ntile.gene.body)
{
  res <- matrix(nrow=length(gene.names),ncol=2*ntile.txs+ntile.gene.body)
  rownames(res) <- gene.names

  bw.file <- import.bw(con = bw.file.name)
  bw.file <- as.data.frame(bw.file)

  for(i in 1:length(gene.names))
  {
    #print(i)
    current.gene.chr <- genes[gene.names[i],"seqnames"]
    current.gene.start <- genes[gene.names[i],"start"]
    current.gene.end <- genes[gene.names[i],"end"]
    current.gene.strand <- genes[gene.names[i],"strand"]

    gene.body.signal <- subset(bw.file, seqnames == current.gene.chr &
                             start >= current.gene.start &
                             end <= current.gene.end)[["score"]]

    upstream.start.signal <- subset(bw.file, seqnames == current.gene.chr &
                                   start >= (current.gene.start - 2000) &
                                   end <= current.gene.start)[["score"]]

    downstream.end.signal <- subset(bw.file, seqnames == current.gene.chr &
                                   start >= current.gene.end &
                                   end <= (current.gene.end + 2000))[["score"]]

    ntile.signal.complete <-
      c(ntile.signal(signal.vector = upstream.start.signal,number.tiles = ntile.txs),
        ntile.signal(signal.vector = gene.body.signal,number.tiles = ntile.gene.body),
        ntile.signal(signal.vector = downstream.end.signal,number.tiles = ntile.txs))

    if(current.gene.strand == "+")
    {
      res[i,] <- ntile.signal.complete
    }
  }
}

```

```

} else
{
  res[i,] <- rev(ntile.signal.complete)
}
}

return(res)
}

## Genes with their entire gene body marked
k27.gene.body <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "gene.body")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.gene.body) <- NULL
length(k27.gene.body)

## [1] 160

## Genes with H3K27 inside their gene body with no overlap
## in the TSS or TES.
k27.internal.gene.body <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "internal.gene.body")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.internal.gene.body) <- NULL
length(k27.internal.gene.body)

## [1] 203

## Genes with H3K27me3 overlapping their TSS
k27.tss <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "TSS")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.tss) <- NULL
length(k27.tss)

## [1] 54

## Genes with H3K27me3 overlapping their TES
k27.tes <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "TES")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.tes) <- NULL
length(k27.tes)

## [1] 74

```

Next we generate the metagenepLOTS for genes where the H3K27me3 occupies their entire gene body including TSS and TES.

```

library(rtracklayer)

## Genes with their entire gene body marked
k27.gene.body <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "gene.body")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.gene.body) <- NULL
length(k27.gene.body)

## [1] 160

## Genes with H3K27 inside their gene body with no overlap
## in the TSS or TES.
k27.internal.gene.body <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "internal.gene.body")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))

```

```
names(k27.internal.gene.body) <- NULL
length(k27.internal.gene.body)
```

```
## [1] 203
```

```
## Genes with H3K27me3 overlapping their TSS
```

```
k27.tss <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "TSS")["target.genes"],
                           FUN = function(x) {strsplit(x, ",")[[1]]})))
```

```
names(k27.tss) <- NULL
length(k27.tss)
```

```
## [1] 54
```

```
## Genes with H3K27me3 overlapping their TES
```

```
k27.tes <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "TES")["target.genes"],
                           FUN = function(x) {strsplit(x, ",")[[1]]})))
```

```
names(k27.tes) <- NULL
length(k27.tes)
```

```
## [1] 74
```

```
metagenenplot.data.k27.gene.body.1 <-
```

```
  metagenenplot(gene.names = k27.gene.body,
                genes = ostta.genes,
                bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_1.bw",
                ntile.txs = 10,
                ntile.gene.body = 20)
```

```
median.metagenenplot.data.k27.gene.body.1 <-
```

```
  apply(X = metagenenplot.data.k27.gene.body.1,
        MARGIN = 2,
        FUN = median, na.rm = T)
```

```
metagenenplot.data.k27.gene.body.2 <-
```

```
  metagenenplot(gene.names = k27.gene.body,
                genes = ostta.genes,
                bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_2n.bw",
                ntile.txs = 10,
                ntile.gene.body = 20)
```

```
median.metagenenplot.data.k27.gene.body.2 <-
```

```
  apply(X = metagenenplot.data.k27.gene.body.2,
        MARGIN = 2,
        FUN = median, na.rm = T)
```

```
metagenenplot.data.k27.gene.body.3 <-
```

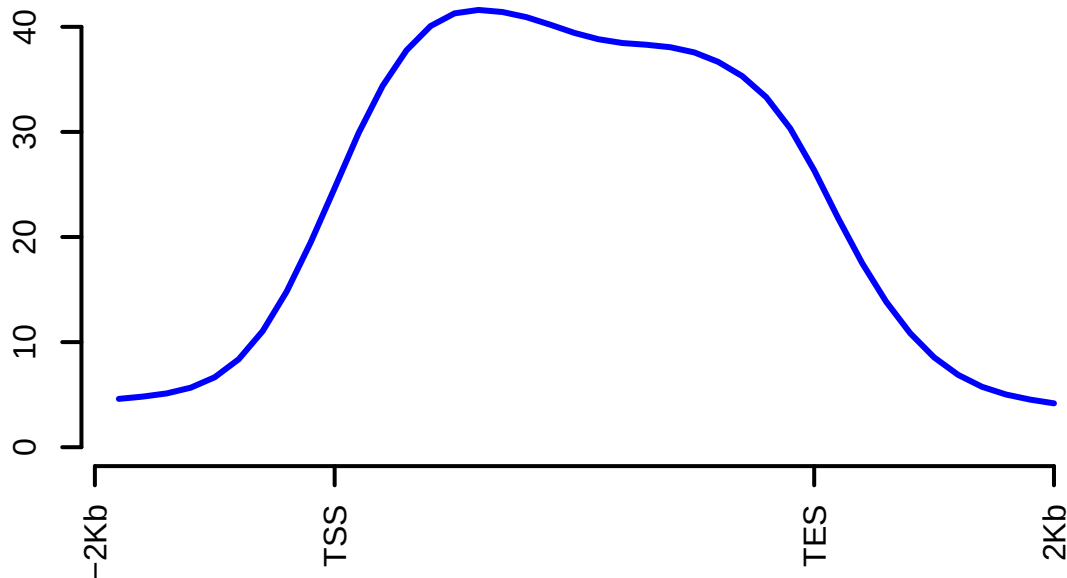
```
  metagenenplot(gene.names = k27.gene.body,
                genes = ostta.genes,
                bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_3n.bw",
                ntile.txs = 10,
                ntile.gene.body = 20)
```

```
median.metagenenplot.data.k27.gene.body.3 <-
```

```
  apply(X = metagenenplot.data.k27.gene.body.3,
        MARGIN = 2,
        FUN = median, na.rm = T)
```

```
median.metagenenplot.data.k27.gene.body <- (median.metagenenplot.data.k27.gene.body.1 + median.metagenenplot.data.k27.gene.body.2 + median.metagenenplot.data.k27.gene.body.3)
```

```
plot(smooth.spline(x=1:40,y=median.metageneplot.data.k27.gene.body,spar = 0.5),
     type="l",col="blue",lwd=3,ylim=c(0,45),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)
```



Next we generate the metageneplots for genes where the H3K27me3 occupies part of the gene body without overlapping with the TSS and TES.

```
metageneplot.data.k27.internal.gene.body.1 <-
  metageneplot(gene.names = k27.internal.gene.body,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_1.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metageneplot.data.k27.internal.gene.body.1 <-
  apply(X = metageneplot.data.k27.internal.gene.body.1,
        MARGIN = 2,
        FUN = median,na.rm = T)

metageneplot.data.k27.internal.gene.body.2 <-
  metageneplot(gene.names = k27.internal.gene.body,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_2n.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metageneplot.data.k27.internal.gene.body.2 <-
  apply(X = metageneplot.data.k27.internal.gene.body.2,
        MARGIN = 2,
        FUN = median,na.rm = T)

metageneplot.data.k27.internal.gene.body.3 <-
  metageneplot(gene.names = k27.internal.gene.body,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_3n.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
```

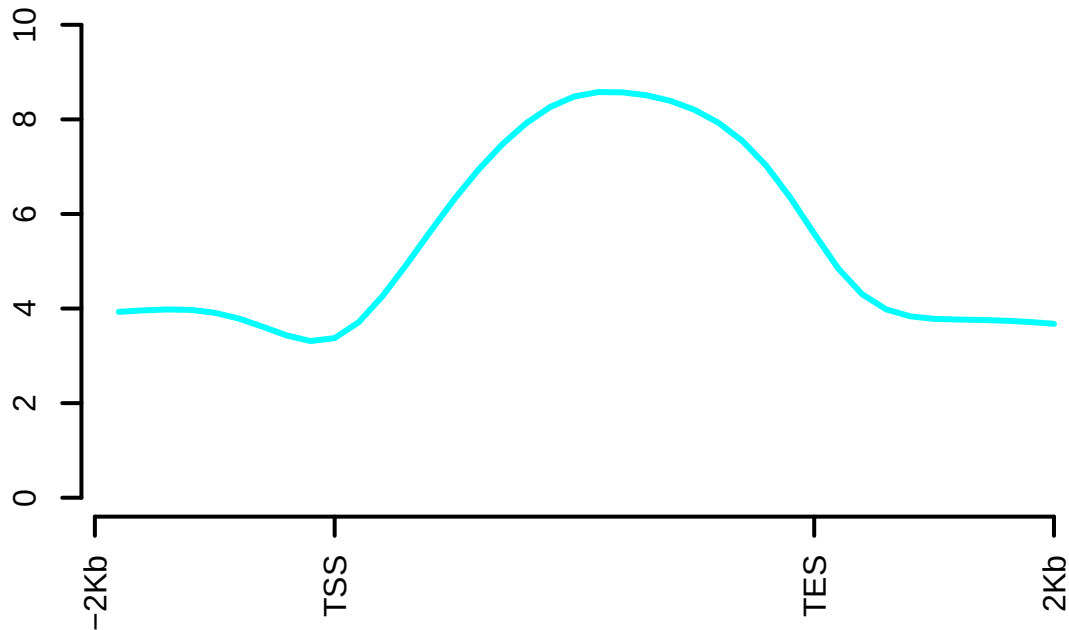
```

median.metagenesplot.data.k27.internal.gene.body.3 <-
  apply(X = metagenesplot.data.k27.internal.gene.body.3,
        MARGIN = 2,
        FUN = median, na.rm = T)

median.metagenesplot.data.k27.internal.gene.body <- (median.metagenesplot.data.k27.internal.gene.body.1 +
  median.metagenesplot.data.k27.internal.gene.body.2 + median.metagenesplot.data.k27.internal.gene.body.3)

plot(smooth.spline(x=1:40,y=median.metagenesplot.data.k27.internal.gene.body,spar = 0.5),
     type="l",col="cyan",lwd=3,ylim=c(0,10),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



Next we generate the metagenesplots for genes where the H3K27me3 occupies only the TSS and the beginning of the gene.

```

metagenesplot.data.k27.tss.1 <-
  metagenesplot(gene.names = k27.tss,
                genes = ostta.genes,
                bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_1.bw",
                ntile.txs = 10,
                ntile.gene.body = 20)
median.metagenesplot.data.k27.tss.1 <-
  apply(X = metagenesplot.data.k27.tss.1,
        MARGIN = 2,
        FUN = median, na.rm = T)

metagenesplot.data.k27.tss.2 <-
  metagenesplot(gene.names = k27.tss,
                genes = ostta.genes,
                bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_2n.bw",
                ntile.txs = 10,
                ntile.gene.body = 20)
median.metagenesplot.data.k27.tss.2 <-

```

```

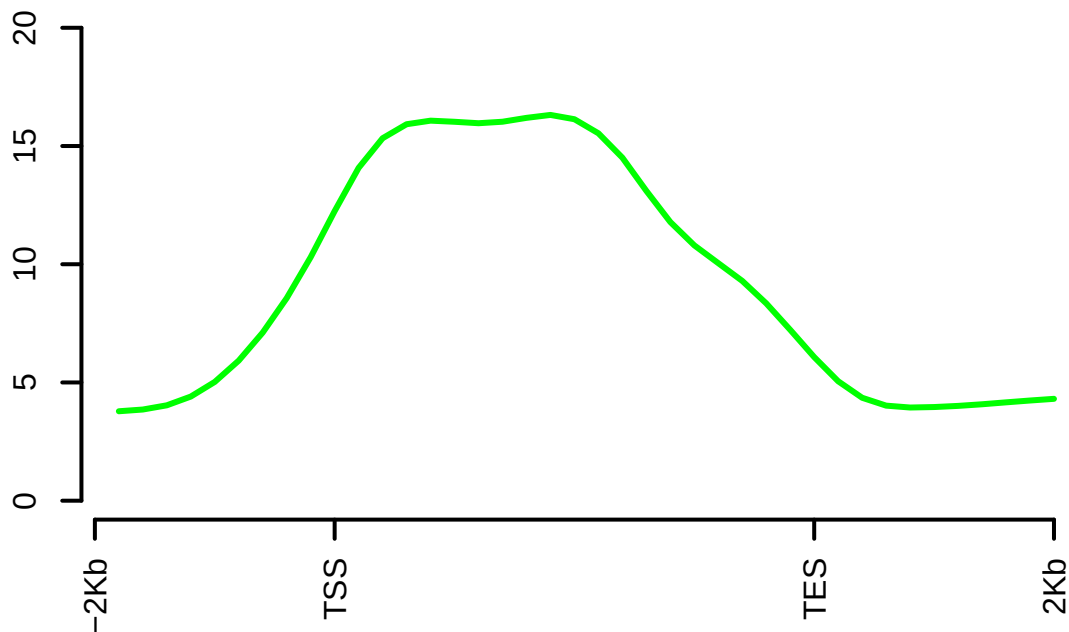
apply(X = metageneplot.data.k27.tss.2,
      MARGIN = 2,
      FUN = median, na.rm = T)

metageneplot.data.k27.tss.3 <-
  metageneplot(gene.names = k27.tss,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_3n.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metageneplot.data.k27.tss.3 <-
  apply(X = metageneplot.data.k27.tss.3,
        MARGIN = 2,
        FUN = median, na.rm = T)

median.metageneplot.data.k27.tss <-
  (median.metageneplot.data.k27.tss.1 +
   median.metageneplot.data.k27.tss.2 +
   median.metageneplot.data.k27.tss.3)

plot(smooth.spline(x=1:40,y=median.metageneplot.data.k27.tss,spar = 0.5),
     type="l",col="green",lwd=3,ylim=c(0,20),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



Next we generate the metageneplots for genes where the H3K27me3 occupies only the TES and the end of the gene.

```

metageneplot.data.k27.tes.1 <-
  metageneplot(gene.names = k27.tes,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_1.bw",
               ntile.txs = 10,

```

```

        ntile.gene.body = 20)
median.metagenepLOT.data.k27.tes.1 <-
  apply(X = metagenepLOT.data.k27.tes.1,
        MARGIN = 2,
        FUN = median,na.rm = T)

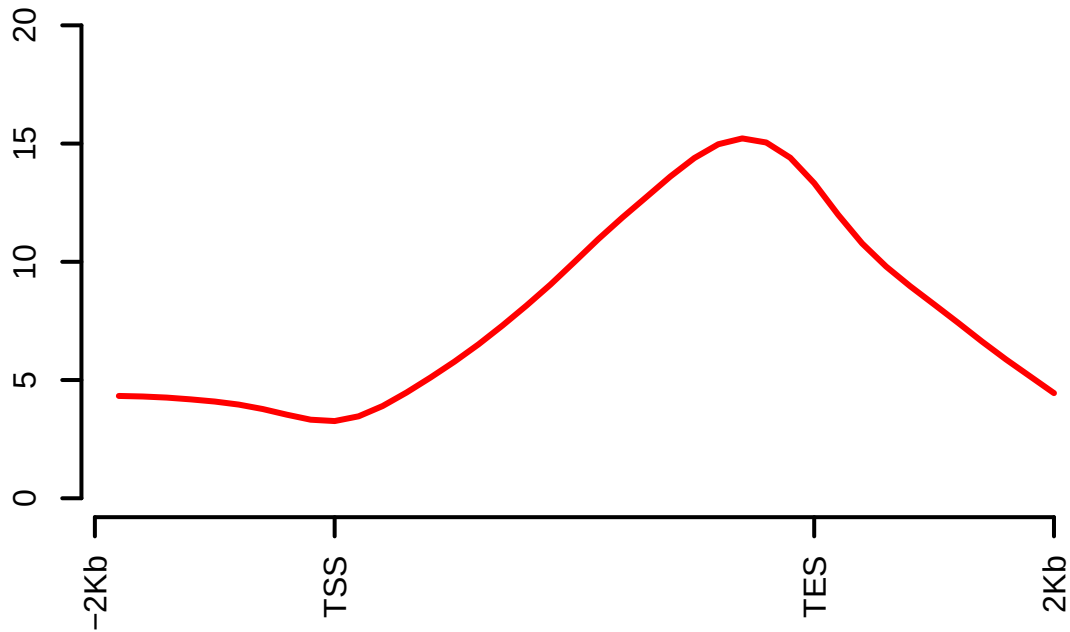
metagenepLOT.data.k27.tes.2 <-
  metagenepLOT(gene.names = k27.tes,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_2n.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k27.tes.2 <-
  apply(X = metagenepLOT.data.k27.tes.2,
        MARGIN = 2,
        FUN = median,na.rm = T)

metagenepLOT.data.k27.tes.3 <-
  metagenepLOT(gene.names = k27.tes,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_3n.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k27.tes.3 <-
  apply(X = metagenepLOT.data.k27.tes.3,
        MARGIN = 2,
        FUN = median,na.rm = T)

median.metagenepLOT.data.k27.tes <-
  (median.metagenepLOT.data.k27.tes.1 +
   median.metagenepLOT.data.k27.tes.2 +
   median.metagenepLOT.data.k27.tes.3)

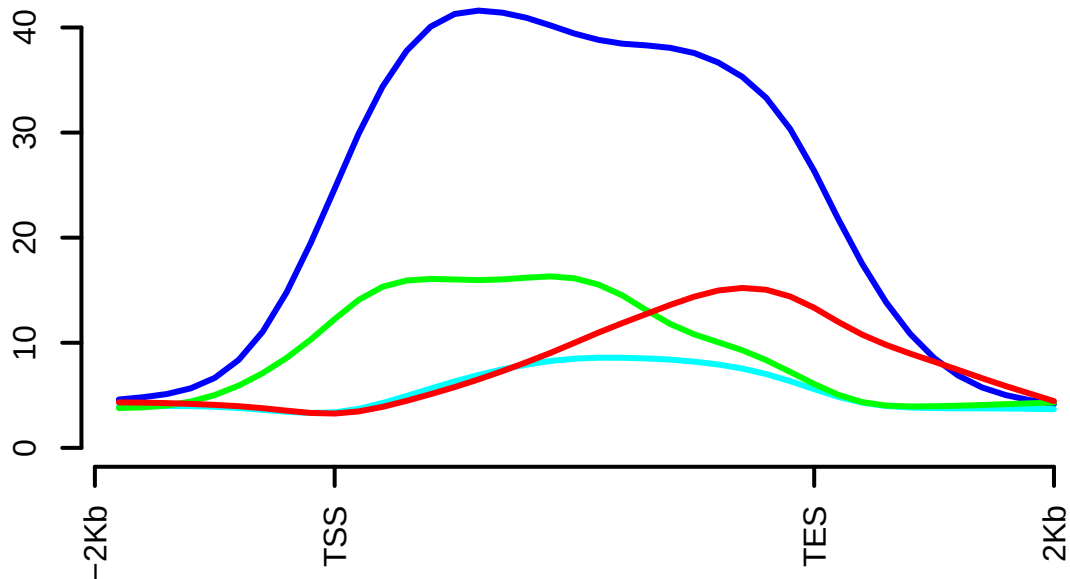
plot(smooth.spline(x=1:40,y=median.metagenepLOT.data.k27.tes,spar = 0.5),
     type="l",col="red",lwd=3,ylim=c(0,20),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



In order to compare the different metagenepoints are represented in the same plot.

```
#jpeg(filename = "images/h3k27me3_metagenepoints.jpg",res = 250,width = 900,height = 900)
plot(smooth.spline(x=1:40,y=median.metagenepoint.data.k27.gene.body,spar = 0.5),
     type="l",col="blue",lwd=3,ylim=c(0,45),axes=F,xlab="",ylab="")
lines(smooth.spline(x=1:40,y=median.metagenepoint.data.k27.internal.gene.body,spar = 0.5),
      type="l",col="cyan",lwd=3)
lines(smooth.spline(x=1:40,y=median.metagenepoint.data.k27.tss,spar = 0.5),
      type="l",col="green",lwd=3)
lines(smooth.spline(x=1:40,y=median.metagenepoint.data.k27.tes,spar = 0.5),
      type="l",col="red",lwd=3)
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb", "TSS", "TES", "2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)
```



```
#dev.off()
```

It can be observed that the level of H3K27me3 mark varies depending on the part of gene that it occupies. Next we check if these differences are significant.

```
max.metagenepplot.data.k27.gene.body.1 <-
  apply(X = metagenepplot.data.k27.gene.body.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.gene.body.2 <-
  apply(X = metagenepplot.data.k27.gene.body.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.gene.body.3 <-
  apply(X = metagenepplot.data.k27.gene.body.3,
        MARGIN = 1,
        FUN = max, na.rm = T)

max.metagenepplot.data.k27.gene.body <- (max.metagenepplot.data.k27.gene.body.1 +
                                          max.metagenepplot.data.k27.gene.body.2 +
                                          max.metagenepplot.data.k27.gene.body.3)/3

max.metagenepplot.data.k27.internal.gene.body.1 <-
  apply(X = metagenepplot.data.k27.internal.gene.body.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.internal.gene.body.2 <-
  apply(X = metagenepplot.data.k27.internal.gene.body.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.internal.gene.body.3 <-
  apply(X = metagenepplot.data.k27.internal.gene.body.3,
        MARGIN = 1,
        FUN = max, na.rm = T)

max.metagenepplot.data.k27.internal.gene.body <- (max.metagenepplot.data.k27.internal.gene.body.1 +
                                                    max.metagenepplot.data.k27.internal.gene.body.2 +
                                                    max.metagenepplot.data.k27.internal.gene.body.3)/3

max.metagenepplot.data.k27.tss.1 <-
  apply(X = metagenepplot.data.k27.tss.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.tss.2 <-
  apply(X = metagenepplot.data.k27.tss.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.tss.3 <-
  apply(X = metagenepplot.data.k27.tss.3,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.tss <- (max.metagenepplot.data.k27.tss.1 +
                                   max.metagenepplot.data.k27.tss.2 +
                                   max.metagenepplot.data.k27.tss.3)/3
```

```

max.metagenepplot.data.k27.tes.1 <-
  apply(X = metagenepplot.data.k27.tes.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.tes.2 <-
  apply(X = metagenepplot.data.k27.tes.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.tes.3 <-
  apply(X = metagenepplot.data.k27.tes.3,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.tes <- (max.metagenepplot.data.k27.tes.1 +
  max.metagenepplot.data.k27.tes.2 +
  max.metagenepplot.data.k27.tes.3)/3

shapiro.test(max.metagenepplot.data.k27.gene.body)

##
##  Shapiro-Wilk normality test
##
## data:  max.metagenepplot.data.k27.gene.body
## W = 0.85058, p-value = 1.774e-11

shapiro.test(max.metagenepplot.data.k27.internal.gene.body)

##
##  Shapiro-Wilk normality test
##
## data:  max.metagenepplot.data.k27.internal.gene.body
## W = 0.55799, p-value < 2.2e-16

shapiro.test(max.metagenepplot.data.k27.tss)

##
##  Shapiro-Wilk normality test
##
## data:  max.metagenepplot.data.k27.tss
## W = 0.92824, p-value = 0.00311

shapiro.test(max.metagenepplot.data.k27.tes)

##
##  Shapiro-Wilk normality test
##
## data:  max.metagenepplot.data.k27.tes
## W = 0.91514, p-value = 0.0001057

wilcox.test(x = max.metagenepplot.data.k27.gene.body,
            y = max.metagenepplot.data.k27.internal.gene.body,
            alternative = "greater")$p.value

## [1] 1.47155e-42

wilcox.test(x = max.metagenepplot.data.k27.gene.body,
            y = max.metagenepplot.data.k27.tss,
            alternative = "greater")

```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenesplot.data.k27.gene.body and max.metagenesplot.data.k27.tss
## W = 6394, p-value = 6.818e-08
## alternative hypothesis: true location shift is greater than 0
```

```
wilcox.test(x = max.metagenesplot.data.k27.gene.body,
            y = max.metagenesplot.data.k27.tss,
            alternative = "greater")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenesplot.data.k27.gene.body and max.metagenesplot.data.k27.tss
## W = 9618, p-value = 8.034e-15
## alternative hypothesis: true location shift is greater than 0
```

```
wilcox.test(x = max.metagenesplot.data.k27.tss,
            y = max.metagenesplot.data.k27.tss)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenesplot.data.k27.tss and max.metagenesplot.data.k27.tss
## W = 2285.5, p-value = 0.1661
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(x = max.metagenesplot.data.k27.tss,
            y = max.metagenesplot.data.k27.tss)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenesplot.data.k27.tss and max.metagenesplot.data.k27.tss
## W = 2285.5, p-value = 0.1661
## alternative hypothesis: true location shift is not equal to 0
```

Indeed the differences between the metagenesplots are highly significant except the ones corresponding to the marks overlapping with the TSS or TES.

Now we analyse the gene expression level of marked genes.

```
temp.gene.expression <- read.table(file="rnaseq_data/ostta_temperature_gene_expression.tsv",header = T,
gene.expression.20 <-
  (temp.gene.expression$T20_1 +
   temp.gene.expression$T20_2 +
   temp.gene.expression$T20_3)/3
names(gene.expression.20) <- rownames(temp.gene.expression)
non.k27.genes <- setdiff(names(gene.expression.20),c(k27.gene.body,k27.tss,k27.tss,k27.internal.gene.bo
summary(gene.expression.20[k27.gene.body])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 0.00425  0.19081  0.90669 12.61720  4.35649 184.25682
```

```
summary(gene.expression.20[k27.internal.gene.body])
```

```
##      Min.   1st Qu.   Median     Mean  3rd Qu.    Max.
##  0.0042   6.1744  14.2076  54.7586  35.0676  785.5262
```

```
summary(gene.expression.20[k27.tss])
```

```
##      Min.   1st Qu.   Median     Mean  3rd Qu.    Max.
##  0.0042   0.4393   2.9986  33.4461  14.0569  566.7411
```

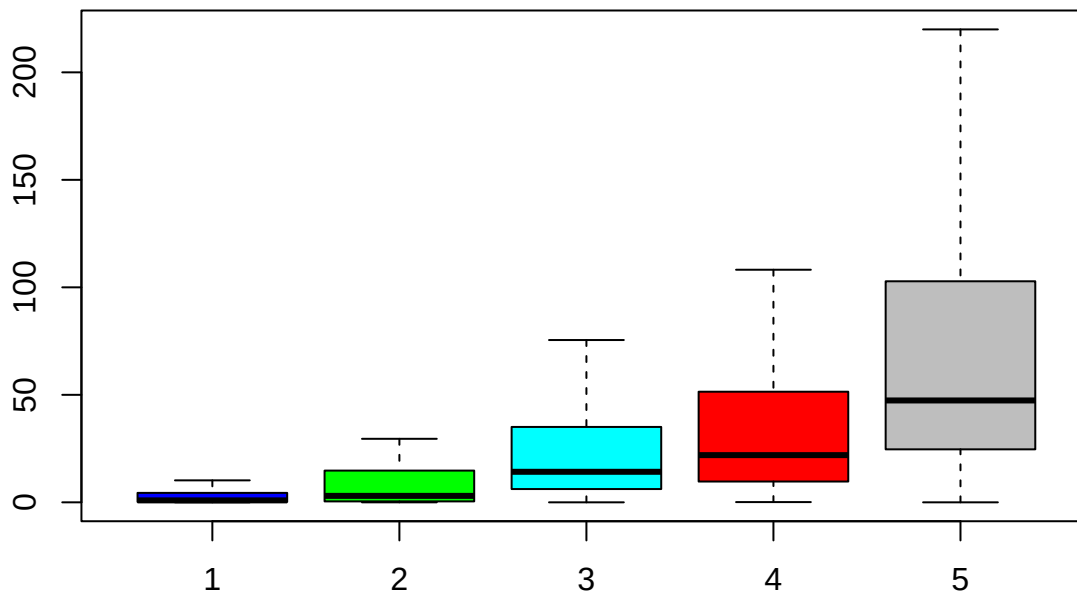
```
summary(gene.expression.20[k27.tes])
```

```
##      Min.   1st Qu.   Median     Mean  3rd Qu.    Max.
##  0.1178   9.8206  21.9533  47.6954  51.0203  297.9537
```

```
summary(gene.expression.20[non.k27.genes])
```

```
##      Min.   1st Qu.   Median     Mean  3rd Qu.    Max.
##  0.004    24.641    47.380   124.142  102.811  13604.318
```

```
#jpeg(filename = "images/h3k27me3_expression_barplot_different_genes.jpg",res = 250,width = 600,height = 400)
boxplot(gene.expression.20[k27.gene.body],
        gene.expression.20[k27.tss],
        gene.expression.20[k27.internal.gene.body],
        gene.expression.20[k27.tes],
        gene.expression.20[non.k27.genes],
        col=c("blue","green","cyan","red","grey"),
        outline=F)
```



```
#dev.off()
```

```
wilcox.test(x = gene.expression.20[k27.gene.body],
            y = gene.expression.20[k27.tss],
            alternative = "less")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
```

```
## data: gene.expression.20[k27.gene.body] and gene.expression.20[k27.tss]
## W = 3352.5, p-value = 0.006989
## alternative hypothesis: true location shift is less than 0
wilcox.test(x = gene.expression.20[k27.tss],
            y = gene.expression.20[k27.internal.gene.body],
            alternative = "less")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k27.tss] and gene.expression.20[k27.internal.gene.body]
## W = 3109.5, p-value = 5.199e-07
## alternative hypothesis: true location shift is less than 0
wilcox.test(x = gene.expression.20[k27.internal.gene.body],
            y = gene.expression.20[k27.tss],
            alternative = "less")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k27.internal.gene.body] and gene.expression.20[k27.tss]
## W = 6295, p-value = 0.01968
## alternative hypothesis: true location shift is less than 0
wilcox.test(x = gene.expression.20[k27.tss],
            y = gene.expression.20[non.k27.genes],
            alternative = "less")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k27.tss] and gene.expression.20[non.k27.genes]
## W = 170128, p-value = 3.801e-08
## alternative hypothesis: true location shift is less than 0
```

We can see all H3K27me3 genes present levels of expression significantly lower than non-marked genes. Specifically, genes whose entire gene body or TSS is occupied by H3K27me3 are completely repressed. Whereas, genes where only the TES or gene body is occupied by H3K27me3 are expressed although at a significantly lower level than non-marked genes.

In what follows we perform a functional analysis over the H3K27me3 marked genes. First, we analyze the relationship between H3K27me3 peaks and transposable elements.

```
dna.transposons <- read.table(file="te_analysis/dna_transposons_unsorted.bed", sep="\t", header=F)
nrow(dna.transposons)
```

```
## [1] 35
```

```
ltr.transposons <- read.table(file="te_analysis/ltr_transposons_unsorted.bed", sep="\t", header=F)
nrow(ltr.transposons)
```

```
## [1] 69
```

```
unclassified.transposons <- read.table(file="te_analysis/unknown_transposons_unsorted.bed", sep="\t", header=F)
nrow(unclassified.transposons)
```

```
## [1] 988
```

```

overlapping.peaks <- function(target.peak, peaks)
{
  subset(peaks, V1 == target.peak[1] & (
    (V2 >= target.peak[2] & V2 <= target.peak[3]) |
    (V3 >= target.peak[2] & V3 <= target.peak[3]) |
    (V2 <= target.peak[2] & V3 >= target.peak[3])))
}

peak.te <- vector(mode = "character", length = nrow(ll.20.peak.annotation))

for(i in 1:nrow(ll.20.peak.annotation))
{
  current.peak <- unlist(ll.20.peak.annotation[i,1:3])

  if(nrow(overlapping.peaks(current.peak,dna.transposons)) > 0)
  {
    peak.te[i] <- "DNA_transposon"
  } else if (nrow(overlapping.peaks(current.peak,ltr.transposons)) > 0)
  {
    peak.te[i] <- "LTR_transposon"
  } else if (nrow(overlapping.peaks(current.peak,unclassified.transposons)) > 0)
  {
    peak.te[i] <- "Unclassified_transposon"
  } else
  {
    peak.te[i] <- "No_transposon"
  }
}

100*table(peak.te)/sum(table(peak.te))

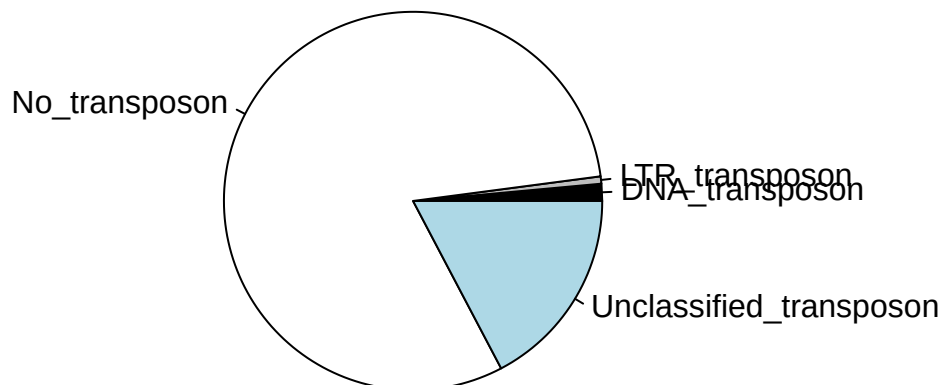
```

```

## peak.te
##          DNA_transposon          LTR_transposon          No_transposon
##             1.458333              0.625000             80.625000
## Unclassified_transposon
##             17.291667

```

pie(table(peak.te), col=c("black","grey","white","lightblue"))



```
nrow(dna.transposons)
```

```

## [1] 35
100*table(peak.te)["DNA_transposon"]/nrow(dna.transposons)

## DNA_transposon
##          20
nrow(ltr.transposons)

## [1] 69
100*table(peak.te)["LTR_transposon"]/nrow(ltr.transposons)

## LTR_transposon
##      4.347826
nrow(unclassified.transposons)

## [1] 988
100*table(peak.te)["Unclassified_transposon"]/nrow(unclassified.transposons)

## Unclassified_transposon
##          8.40081
table(ll.20.peak.annotation$peak.annotation[peak.te == "DNA_transposon"])

##
##      gene.body      intergenic internal.gene.body
##          2          4          1
table(ll.20.peak.annotation$peak.annotation[peak.te == "LTR_transposon"])

##
##  gene.body intergenic
##      1      2
table(ll.20.peak.annotation$peak.annotation[peak.te == "Unclassified_transposon"])

##
##      gene.body      intergenic internal.gene.body      TES
##          24          25          14          13
##      TSS
##          7
table(ll.20.peak.annotation$peak.annotation)

##
##      gene.body      intergenic internal.gene.body      TES
##          114          38          201          74
##      TSS
##          53
(table(ll.20.peak.annotation$peak.annotation[peak.te == "DNA_transposon"])["intergenic"] +
table(ll.20.peak.annotation$peak.annotation[peak.te == "LTR_transposon"])["intergenic"] +
table(ll.20.peak.annotation$peak.annotation[peak.te == "Unclassified_transposon"])["intergenic"]) / tab

## intergenic
## 0.8157895
ll.20.peak.annotation <- cbind(ll.20.peak.annotation,peak.te)
write.table(x = ll.20.peak.annotation,file = "tables/H3K27me3_LL_20C_peak_annotation.tsv",quote = F,sep

```

H3K27me3 is only partially associated with Transposable Elements. More than 81% of the peaks did not overlap with any TE. Only 1.5%, 0.6% and 17.3% of the H3K27me3 peaks were found overlapping with DNA, LTR and unclassified transposons respectively. Most TE were neither associated with H3K27me3 peaks, specifically 7 out of 35 (20%) DNA transposons, 3 out of 69 (4.3%) LTR transposons and 83 out of 988 unclassified transposons were H3K27me3 marked. Most H3K27me3 marks located at intergenic regions (81.6%) were associated with transposons

```
library(clusterProfiler)

##

## clusterProfiler v4.4.4 For help: https://yulab-smu.top/biomedical-knowledge-mining-book/
##
## If you use clusterProfiler in published research, please cite:
## T Wu, E Hu, S Xu, M Chen, P Guo, Z Dai, T Feng, L Zhou, W Tang, L Zhan, X Fu, S Liu, X Bo, and G Yu.
##
## Attaching package: 'clusterProfiler'

## The following object is masked from 'package:AnnotationDbi':
##
##     select

## The following object is masked from 'package:IRanges':
##
##     slice

## The following object is masked from 'package:S4Vectors':
##
##     rename

## The following object is masked from 'package:stats':
##
##     filter

library(enrichplot)
library(org.Obtativ5.eg.db)

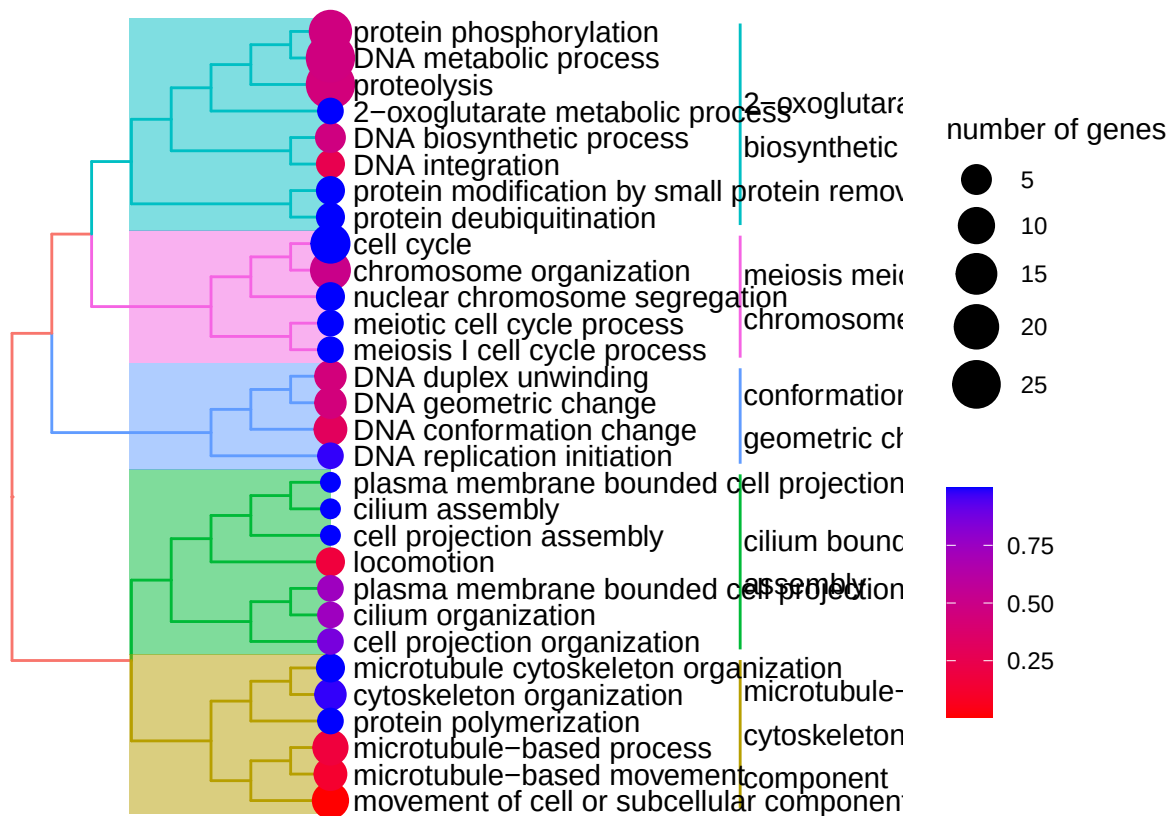
##

k27.enrichment <- enrichGO(gene = all.k27.genes, pvalueCutoff = 0.05, qvalueCutoff = 0.05,
                           OrgDb = org.Obtativ5.eg.db,
                           ont = "BP",
                           keyType = "GID")
k27.enrichment.df <- as.data.frame(k27.enrichment)

k27.go.terms <- enrichGO(gene = all.k27.genes, pvalueCutoff = 1, qvalueCutoff = 1,
                         OrgDb = org.Obtativ5.eg.db,
                         ont = "BP",
                         keyType = "GID")
k27.go.terms.df <- as.data.frame(k27.go.terms)

write.table(x = k27.go.terms.df, file = "go_enrichment_k27.tsv", quote = F, sep = "\t", row.names = F)

treemap(x = pairwise_termsim(k27.go.terms))
```



```
#jpeg(filename = "images/h3k27me3_revigo_treemap.jpg",res = 300,width = 1400,height = 900)
```

```
library(treemap)
```

```
revigo.names <- c("term_ID","description","frequency","value","uniqueness","dispensability","representa  
revigo.data <- rbind(c("GO:0000003","reproduction",1.1154428097959028,0.35455656429893245,1,-0,"reprodu  
c("GO:0001887","selenium compound metabolic process",0.0031524026730797234,0.33700  
c("GO:0002376","immune system process",0.9427113541805001,0.3647586988245658,1,-0,  
c("GO:0006099","tricarboxylic acid cycle",0.5040640114710979,0.40392617241762396,0  
c("GO:0006103","2-oxoglutarate metabolic process",0.017201421622692253,1.024471530  
c("GO:0006525","arginine metabolic process",0.425056765430777,0.6822743332528265,0  
c("GO:0006526","arginine biosynthetic process",0.2840884163423315,0.68227433325282  
c("GO:0006631","fatty acid metabolic process",1.936945636803579,0.661709121736741,  
c("GO:0019344","cysteine biosynthetic process",0.13937119746048948,0.3647586988245  
c("GO:0019541","propionate metabolic process",0.06516280052453614,0.31217797104467  
c("GO:0030258","lipid modification",0.7621839254181456,0.3121779710446768,0.907289  
c("GO:0042128","nitrate assimilation",0.06915814433459262,0.39602842215015766,0.87  
c("GO:0043648","dicarboxylic acid metabolic process",1.2968131796952338,0.34085777  
c("GO:0046487","glyoxylate metabolic process",0.1082932886998466,0.424318332665350  
c("GO:0007017","microtubule-based process",1.4522840599239462,2.9055417594493695,0  
c("GO:0007018","microtubule-based movement",0.6202173565622278,3.3618830741865118,  
c("GO:0007049","cell cycle",2.8073119376140743,1.055588520937294,0.983482157837816  
c("GO:0007154","cell communication",9.671968224175414,0.47456381848456913,0.980753  
c("GO:0009267","cellular response to starvation",0.27913921808780134,0.92702796603  
c("GO:0000160","phosphorelay signal transduction system",2.0739975137672992,0.6100  
c("GO:0007166","cell surface receptor signaling pathway",1.7699816731780171,0.5778  
c("GO:0071496","cellular response to external stimulus",0.6305470826004429,0.72291  
c("GO:0015074","DNA integration",0.7252350055806647,2.657762206770456,0.8035461550
```

c("GO:0000271", "polysaccharide biosynthetic process", 0.8445801891875397, 0.42431833,
c("GO:0005975", "carbohydrate metabolic process", 5.3400641443698875, 0.6489525204320,
c("GO:0006259", "DNA metabolic process", 5.6268046211314795, 2.0741453286090206, 0.788,
c("GO:0006270", "DNA replication initiation", 0.18677061560434888, 1.524614033979224, 0,
c("GO:0006281", "DNA repair", 2.895534847137308, 0.7716708877524712, 0.715963904401935,
c("GO:0006298", "mismatch repair", 0.21782141222315915, 0.6187036479778791, 0.77582612,
c("GO:0006468", "protein phosphorylation", 1.0150465485898383, 2.0478177071574097, 0.8,
c("GO:0006486", "protein glycosylation", 0.7526798873357411, 0.6623793059324244, 0.770,
c("GO:0006508", "proteolysis", 5.2622572267907, 2.215841795989129, 0.8331839915576025, 0,
c("GO:0006637", "acyl-CoA metabolic process", 0.5539971825062208, 0.39602842215015766,
c("GO:0008213", "protein alkylation", 0.000579213939150692, 0.33320452706495884, 0.894,
c("GO:0009100", "glycoprotein metabolic process", 0.9570906483124367, 0.6014649273302,
c("GO:0009119", "ribonucleoside metabolic process", 0.3506733719429951, 0.396028422150,
c("GO:0009226", "nucleotide-sugar biosynthetic process", 0.35550672740969513, 0.30331,
c("GO:0010501", "RNA secondary structure unwinding", 0.0022996025754365773, 0.3370010,
c("GO:0016266", "O-glycan processing", 0.009247705105078281, 0.6822743332528265, 0.832,
c("GO:0016310", "phosphorylation", 5.235381700014107, 0.9200287559031499, 0.8995173712,
c("GO:0016579", "protein deubiquitination", 0.0016070105886223452, 1.363169883748535, 0,
c("GO:0018202", "peptidyl-histidine modification", 1.0281836136778093, 0.927027966035,
c("GO:0018342", "protein prenylation", 0.00018239077232830298, 0.3647586988245658, 0.8,
c("GO:0019439", "aromatic compound catabolic process", 2.2395984760807317, 0.48842621,
c("GO:0032543", "mitochondrial translation", 0.2221051306327077, 0.7671257790005732, 0,
c("GO:0034404", "nucleobase-containing small molecule biosynthetic process", 0.25803,
c("GO:0034655", "nucleobase-containing compound catabolic process", 1.65646806480531,
c("GO:0035383", "thioester metabolic process", 0.5539971825062208, 0.39602842215015766,
c("GO:0042398", "cellular modified amino acid biosynthetic process", 0.5343877097407,
c("GO:0044270", "cellular nitrogen compound catabolic process", 2.004743244566209, 0.4,
c("GO:0046434", "organophosphate catabolic process", 0.9710632603168847, 0.8684573120,
c("GO:0046700", "heterocycle catabolic process", 2.042108705491845, 0.468500275056522,
c("GO:0070646", "protein modification by small protein removal", 0.0641152859537317, 0,
c("GO:0071897", "DNA biosynthetic process", 0.7667855953091854, 2.0080982132635268, 0.1,
c("GO:0090304", "nucleic acid metabolic process", 13.541109943481535, 0.5761569684631,
c("GO:0140053", "mitochondrial gene expression", 0.3272164397774754, 0.72291622303564,
c("GO:1901361", "organic cyclic compound catabolic process", 2.407836710372425, 0.458,
c("GO:0016441", "regulation of gene expression", 0.049918382594123255, 0.7229162230356,
c("GO:0006450", "regulation of translational fidelity", 0.29561600610151356, 0.364758,
c("GO:0010608", "post-transcriptional regulation of gene expression", 1.681196802896,
c("GO:0010629", "negative regulation of gene expression", 1.1219250764340574, 0.53109,
c("GO:0017148", "negative regulation of translation", 0.21779183534115995, 0.33700103,
c("GO:0031326", "regulation of cellular biosynthetic process", 14.068660464000152, 0.1,
c("GO:0033044", "regulation of chromosome organization", 0.18193972487781543, 0.31217,
c("GO:0034249", "negative regulation of amide metabolic process", 0.2399621731397151,
c("GO:1903311", "regulation of mRNA metabolic process", 0.4779772015478174, 0.3647586,
c("GO:2000241", "regulation of reproductive process", 0.1639520511419782, 0.312177971,
c("GO:0030029", "actin filament-based process", 0.8246601591610891, 0.6822743332528266,
c("GO:0030154", "cell differentiation", 2.279586420543629, 0.4717135495803034, 0.95485,
c("GO:0009888", "tissue development", 0.6842315881691289, 0.3494179598129715, 0.972396,
c("GO:0048229", "gametophyte development", 0.017253181166190824, 0.3647586988245658, 0,
c("GO:0048507", "meristem development", 0.03934957675974807, 0.33700103077893184, 0.97,
c("GO:0040011", "locomotion", 0.5148398554794674, 2.9956162519741056, 1, -0, "locomotion",
c("GO:0043547", "positive regulation of GTPase activity", 0.0008897712001421268, 0.92,
c("GO:0051168", "nuclear export", 0.36225518598584233, 0.8822380235820269, 0.919314523,
c("GO:0006403", "RNA localization", 0.38503431460554743, 0.5719097997394266, 0.9492853,

```

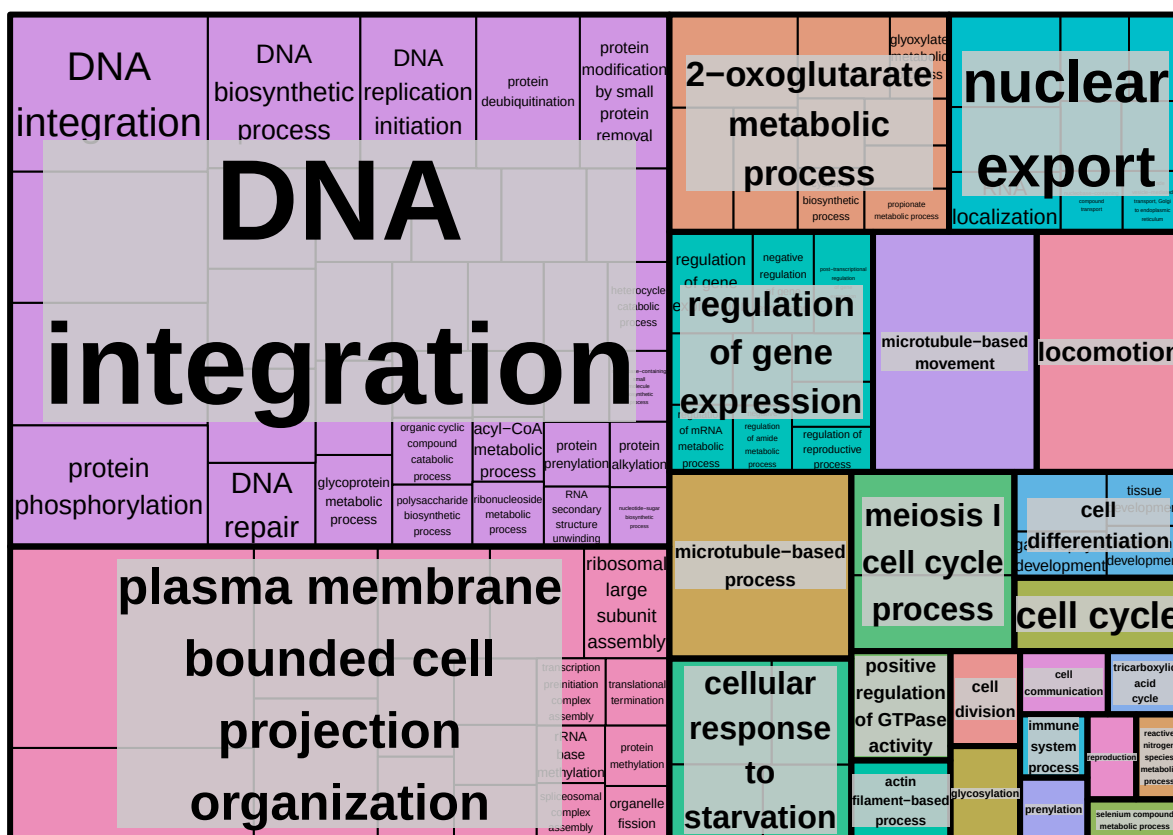
c("GO:0006614", "SRP-dependent cotranslational protein targeting to membrane", 0.173),
c("GO:0006890", "retrograde vesicle-mediated transport, Golgi to endoplasmic reticulum", 0.173),
c("GO:0015931", "nucleobase-containing compound transport", 0.5729682875685309, 0.324),
c("GO:0031503", "protein-containing complex localization", 0.2928998624379218, 0.3647),
c("GO:0033750", "ribosome localization", 0.08453565823400183, 0.39602842215015766, 0.9),
c("GO:0050658", "RNA transport", 0.3380539022900098, 0.6187036479778791, 0.93297221818),
c("GO:0051169", "nuclear transport", 0.5733749696960196, 0.5066832555406448, 0.9263142),
c("GO:0051301", "cell division", 1.5693197819947182, 0.5659276631444706, 0.98451372698),
c("GO:0061982", "meiosis I cell cycle process", 0.22371953544182982, 1.44364678582089),
c("GO:0098813", "nuclear chromosome segregation", 0.33031461816688995, 1.057639560084),
c("GO:0070085", "glycosylation", 0.8372204750500761, 0.5642517085182833, 0.96031272588),
c("GO:0071103", "plasma membrane bounded cell projection organization", 0.8233982121),
c("GO:0000027", "ribosomal large subunit assembly", 0.012821578346646382, 0.815425042),
c("GO:0000245", "spliceosomal complex assembly", 0.1148766096848317, 0.33700103077893),
c("GO:0000280", "nuclear division", 0.45164391760787703, 0.4717135495803034, 0.8236095),
c("GO:0006415", "translational termination", 0.15480047090339727, 0.3647586988245658, 0.8),
c("GO:0006479", "protein methylation", 0.000579213939150692, 0.33320452706495884, 0.89),
c("GO:0006996", "organelle organization", 6.781836087483291, 0.901501456180904, 0.8241),
c("GO:0007010", "cytoskeleton organization", 1.98151060375585, 1.5080424402092374, 0.8),
c("GO:0030030", "cell projection organization", 1.3282977705833654, 1.613362597141604),
c("GO:0042255", "ribosome assembly", 0.2624923630025938, 0.42431833266535085, 0.817136),
c("GO:0043248", "proteasome assembly", 0.04973352708162835, 0.39602842215015766, 0.861),
c("GO:0048285", "organelle fission", 0.5487596096521986, 0.30148135432836, 0.835687575),
c("GO:0051258", "protein polymerization", 0.1780454354145895, 1.0725168728382306, 0.84),
c("GO:0051276", "chromosome organization", 1.5299726699751368, 1.9364166059501378, 0.8),
c("GO:0070475", "rRNA base methylation", 0.22436283262531206, 0.33700103077893184, 0.6),
c("GO:0070897", "transcription preinitiation complex assembly", 0.11026754557329214, 0.8),
c("GO:0070925", "organelle assembly", 1.3561320812847646, 0.6413546437511954, 0.801071),
c("GO:0097435", "supramolecular fiber organization", 0.801568008540226, 0.54778385514),
c("GO:0120036", "plasma membrane bounded cell projection organization", 1.1030821378),
c("GO:0097354", "prenylation", 0.009942761832059112, 0.3647586988245658, 0.94970057450),
c("GO:2001057", "reactive nitrogen species metabolic process", 0.11012952012396263, 0.8)

stuff <- data.frame(revigo.data);
names(stuff) <- revigo.names;

stuff$value <- as.numeric( as.character(stuff$value) );
stuff$frequency <- as.numeric( as.character(stuff$frequency) );
stuff$uniqueness <- as.numeric( as.character(stuff$uniqueness) );
stuff$dispensability <- as.numeric( as.character(stuff$dispensability) );

treemap(
  stuff,
  index = c("representative", "description"),
  vSize = "value",
  type = "categorical",
  vColor = "representative",
  title = "",
  inflate.labels = TRUE,
  lowerbound.cex.labels = 0,
  position.legend = "none"
)

```



```
#dev.off()
```

We next explore the association between ChIP signal and gene expression.

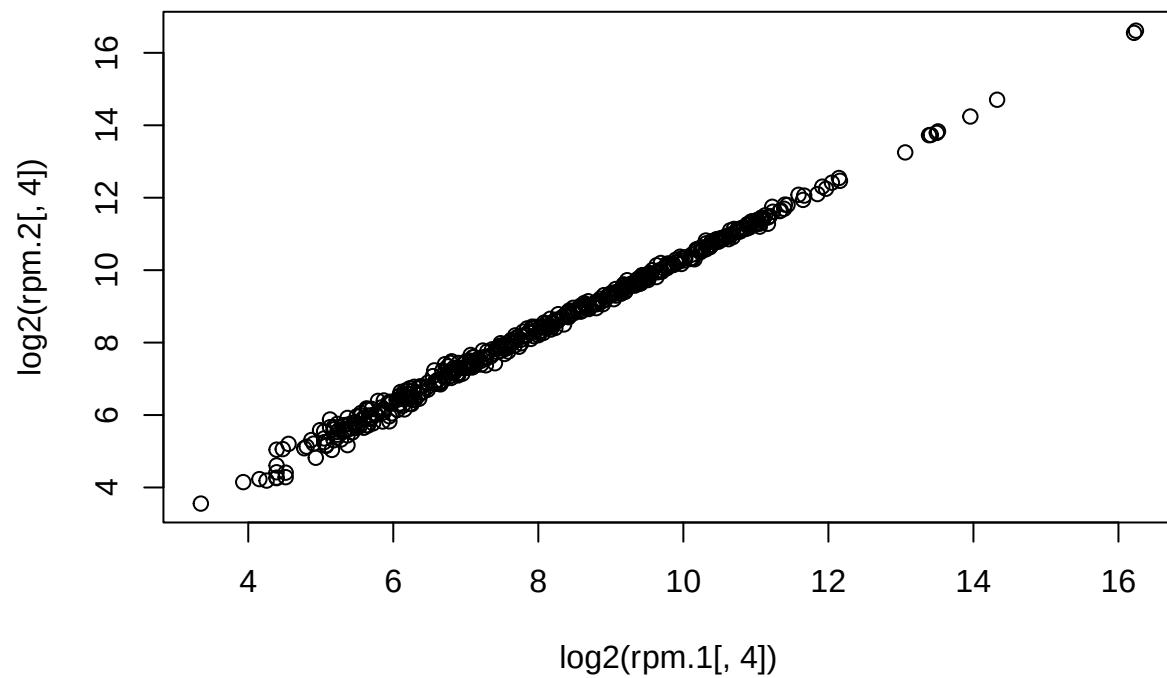
```
count.1 <- read.table(file = "chipseq_data/bed_data/h3k27me3/temp20c/k27_20_1l_chip_1_counts.bed", header=1)
count.2 <- read.table(file="chipseq_data/bed_data/h3k27me3/temp20c/k27_20_1l_chip_2n_counts.bed", header=1)
count.3 <- read.table(file="chipseq_data/bed_data/h3k27me3/temp20c/k27_20_1l_chip_3n_counts.bed", header=1)
count.1 <- count.1[,c(1:3,10)]
count.2 <- count.2[,c(1:3,10)]
count.3 <- count.3[,c(1:3,10)]

mapped.reads.k27.20c.1 <- 3146662/1E6
mapped.reads.k27.20c.2 <- 5697849/1E6
mapped.reads.k27.20c.3 <- 5157582/1E6

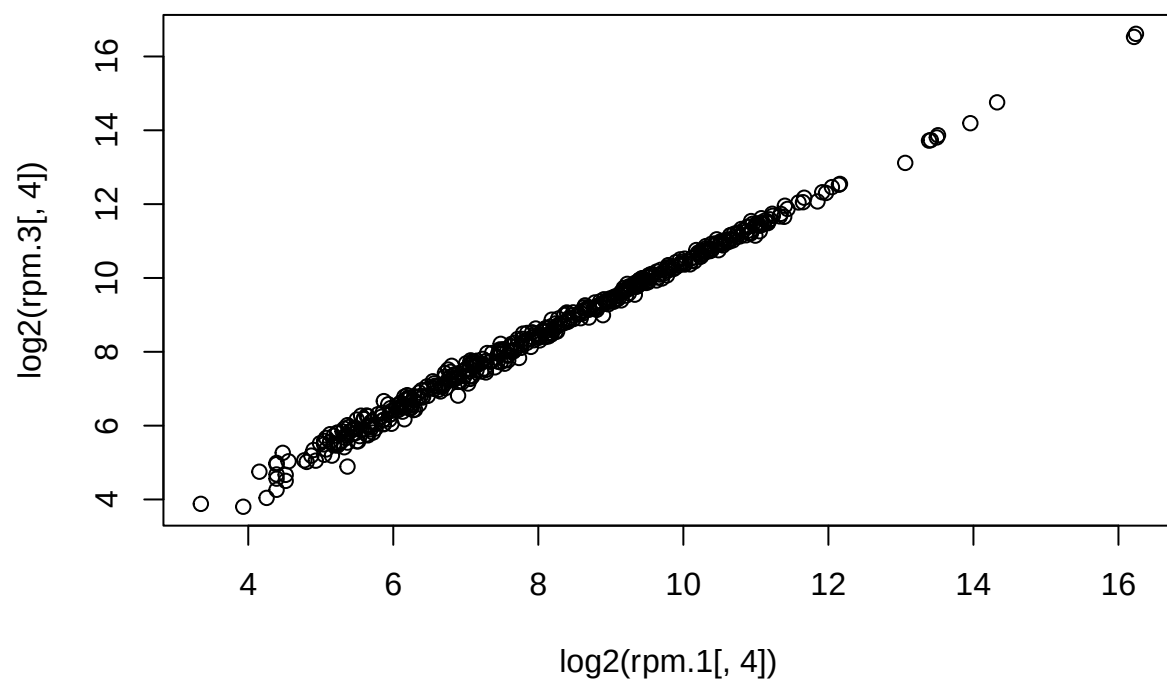
rpm.1 <- count.1
rpm.2 <- count.2
rpm.3 <- count.3

rpm.1[,4] <- rpm.1[,4]/mapped.reads.k27.20c.1
rpm.2[,4] <- rpm.2[,4]/mapped.reads.k27.20c.2
rpm.3[,4] <- rpm.3[,4]/mapped.reads.k27.20c.3

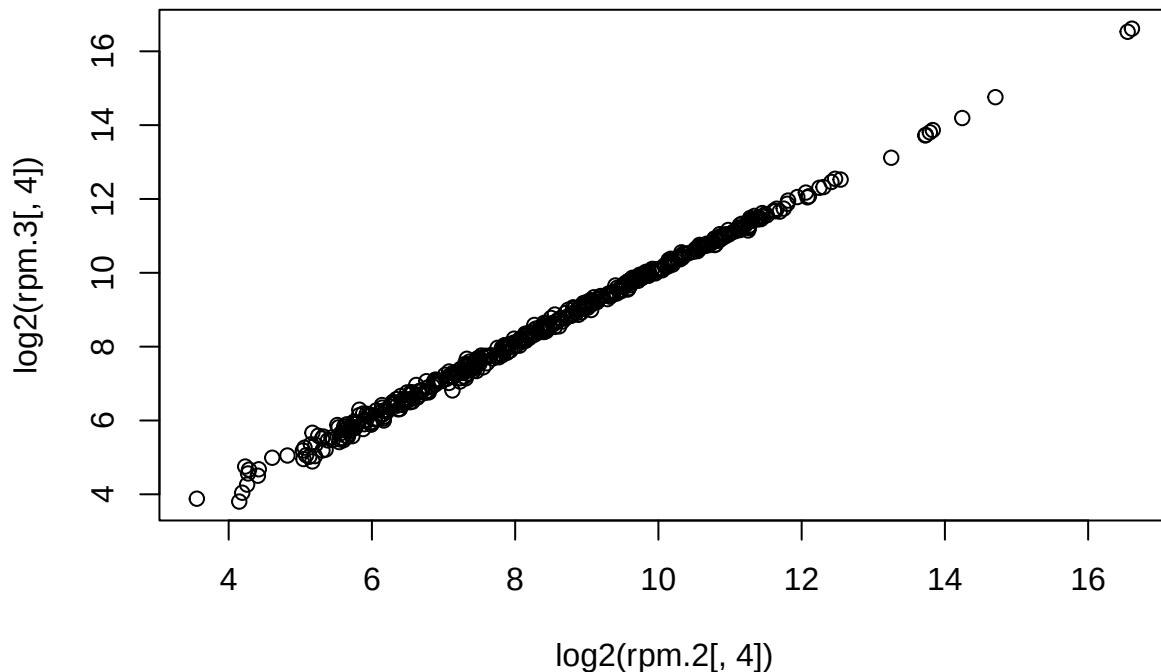
plot(log2(rpm.1[,4]),log2(rpm.2[,4]))
```



```
plot(log2(rpm.1[,4]),log2(rpm.3[,4]))
```



```
plot(log2(rpm.2[,4]),log2(rpm.3[,4]))
```



```
rpm <- rpm.1
rpm[,4] <- (rpm.1[,4] + rpm.2[,4] + rpm.3[,4])/3
```

H3K4me3 genome wide distribution analysis

We load the H3K4me3 peaks identified in cultures grown under constant light at 20°C. We represent each chromosome using a rectangle. Each H3k4me3 peak is represented in its precise genome location using a dark orange rectangle.

```
k4.ll.20 <- read.table(file="chipseq_data/bed_data/h3k4me3/h3k4me3_ll_20.bed",
                      header = F)

k4.ll.20 <- k4.ll.20[,1:3]

nrow(k4.ll.20)

## [1] 773

#jpeg(filename = "images/h3k27me3_genome_wide_distribution_ll_20.jpg",res = 300,width = 900,height = 900)
#par(mar=c(0,0,0,0))
plot(x=c(0,max.chr.len), y=c(0,4*number.chrs), col = "white", xlab = "", ylab = "", axes=F)

for(i in 1:number.chrs)
{
  polygon(x = c(0, chr.lens[i], chr.lens[i], 0),
        y = c(4*i+1, 4*i+1, 4*i-1, 4*i-1),
        col = "white")
}

for(i in 1:nrow(k4.ll.20))
{
  current.chr <- k4.ll.20[i,1]
  current.start <- k4.ll.20[i,2]
```

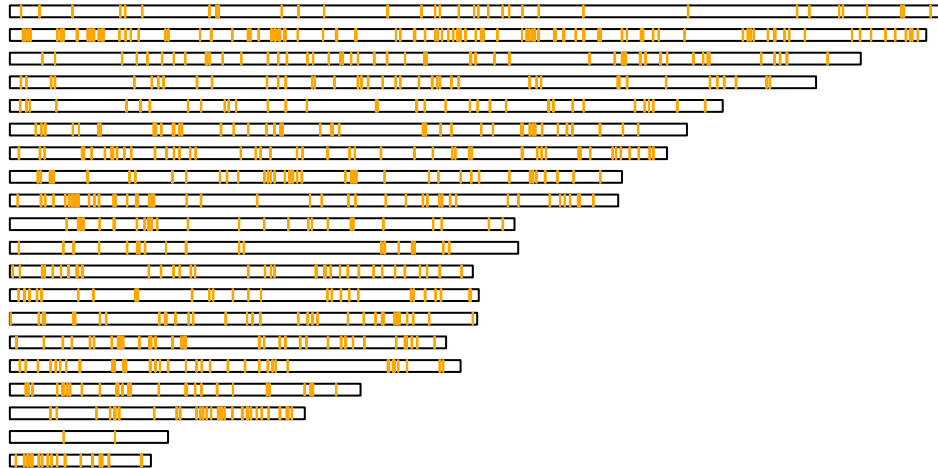
```

current.end <- k4.ll.20[i,3]

current.line <- (20:1)[current.chr]

polygon(x = c(current.start, current.end, current.end, current.start),
        y = c(4*current.line+1, 4*current.line+1, 4*current.line-1, 4*current.line-1),
        col = "orange",border="orange")
}

```



```
#dev.off()
```

We identified 773 genomic regions significantly occupied by H3K4me3 in the three different independent biological replicates. We compute the percentage of the whole genome occupied by H3K4me3 and the percentage in each chromosome. A barplot is used to represent the percentage of H3K4me3 in each chromosome. Z-scores and p-values are computed for each chromosome. A gradient from white to dark orange is used to represent significance levels.

```

h3k4me3.peaks.len <- k4.ll.20$V3 - k4.ll.20$V2
sum(h3k4me3.peaks.len)/sum(chr.len)

```

```
## [1] 0.03710763
```

```

k4.ll.20 <- cbind(k4.ll.20,h3k4me3.peaks.len)
colnames(k4.ll.20) <- c("chr","start","end","length")
head(k4.ll.20)

```

```

##   chr  start    end length
## 1   1  12890  13380    490
## 2   1  34365  35163    798
## 3   1  73034  73837    803
## 4   1 129146 129876    730
## 5   1 134797 135611    814
## 6   1 154932 155506    574

```

```

number.of.h3k4.peaks.chr <- vector(mode = "numeric",length = 20)
k4.peaks.len.chr <- vector(mode = "numeric",length = 20)

```

```

for(i in 1:20)
{
  number.of.h3k4.peaks.chr[i] <- nrow(subset(k4.ll.20, chr == i))
  k4.peaks.len.chr[i] <- sum(subset(k4.ll.20, chr == i)[,4])
}

```

```

}

percentage.chr.k4 <- k4.peaks.len.chr/rev(chr.lens)

k4.z.scores <- (percentage.chr.k4 - mean(percentage.chr.k4))/sd(percentage.chr.k4)

which(k4.z.scores > 2.5)

## [1] 20

which(k4.z.scores < -2.5)

## integer(0)

p.vals <- pnorm((percentage.chr.k4 - mean(percentage.chr.k4[-20]))/sd(percentage.chr.k4[-20]),lower.tail=FALSE)

p.vals

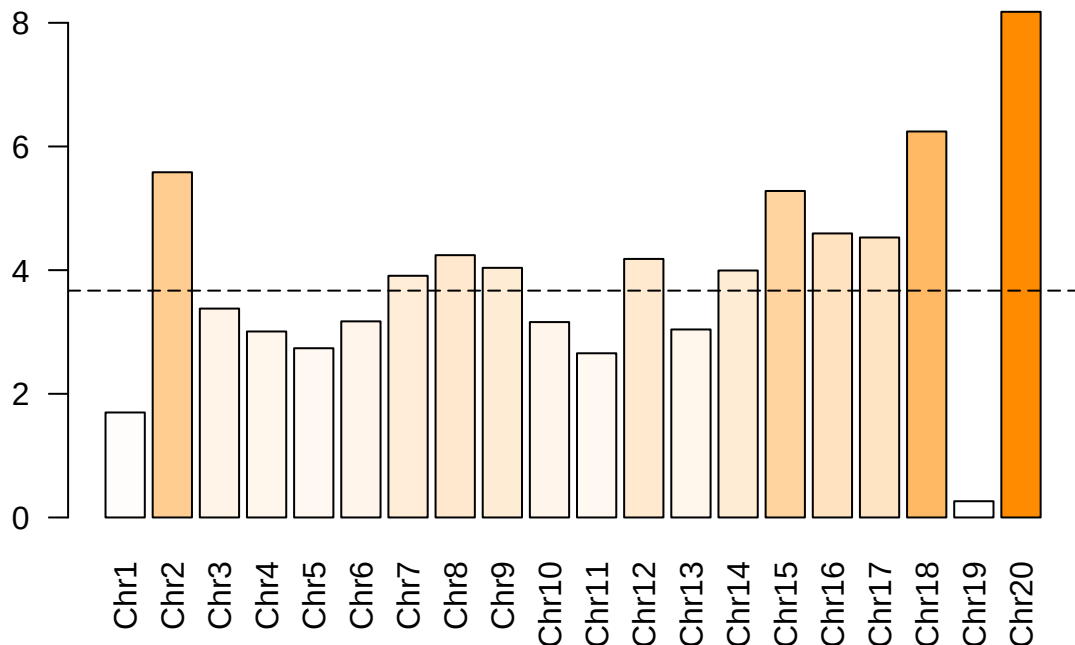
## [1] 0.9236798154 0.0823573562 0.5834699959 0.6842433270 0.7505201875
## [6] 0.6408287755 0.4307406816 0.3385907732 0.3944723502 0.6439650945
## [11] 0.7688748702 0.3549243868 0.6756907811 0.4065474714 0.1210512162
## [16] 0.2510147546 0.2663780211 0.0308763307 0.9932807443 0.0005320007

values.for.colors <- round(-log10(p.vals)*40+1)
values.for.colors[20] <- 100

white.darkorange.palette <- colorRampPalette(c('white','darkorange'))

#jpeg(filename = "images/h3k27me3_chr_barplot_ll_20.jpg",res = 300,width = 1800,height = 900)
barplot(100*percentage.chr.k4,border="black",names.arg = paste0("Chr",1:20),las=2,col=white.darkorange.palette[values.for.colors,])
abline(h = 100*mean(percentage.chr.k4[-20]),lty=5)

```



```
#dev.off()
```

Approximately 3.7% of the *Ostreococcus* genome is occupied by H3K4me3. H3K4me3 is mostly uniformly

distributed over the different chromosome although it significantly accumulates (p-value < 0.01) in Chr20 with p-value 5.32×10^{-4} .

We determine the gene features occupied by each H3K4me3 mark and the H3K4me3 marked genes.

```
library(TxDb.Otauri.JGI)
txdb <- TxDb.Otauri.JGI

ostta.genes <- as.data.frame(genes(txdb))
head(ostta.genes)

##          seqnames start   end width strand      gene_id
## ostta01g00010      1    244  1932  1689      - ostta01g00010
## ostta01g00020      1   2077  4029  1953      + ostta01g00020
## ostta01g00030      1   4157  5299  1143      + ostta01g00030
## ostta01g00040      1   5392  7041  1650      - ostta01g00040
## ostta01g00050      1   7386  9605  2220      - ostta01g00050
## ostta01g00060      1   9780 10412   633      - ostta01g00060

target.genes <- vector(mode = "character",length = nrow(k4.ll.20))
peak.annotation <- vector(mode = "character",length = nrow(k4.ll.20))

for(i in 1:nrow(k4.ll.20))
{
  current.chr <- k4.ll.20[i,1]
  current.start <- k4.ll.20[i,2]
  current.end <- k4.ll.20[i,3]

  complete.gene.body <- subset(ostta.genes,
                              seqnames == current.chr &
                              start >= current.start &
                              end <= current.end)

  overlap.tss.pos.strand <- subset(ostta.genes,
                                  seqnames == current.chr &
                                  start >= current.start & start <= current.end &
                                  strand == "+")

  overlap.tss.neg.strand <- subset(ostta.genes,
                                  seqnames == current.chr &
                                  end >= current.start & end <= current.end &
                                  strand == "-")

  overlap.tes.pos.strand <- subset(ostta.genes,
                                  seqnames == current.chr &
                                  end >= current.start & end <= current.end &
                                  strand == "+")

  overlap.tes.neg.strand <- subset(ostta.genes,
                                  seqnames == current.chr &
                                  start >= current.start & start <= current.end &
                                  strand == "-")

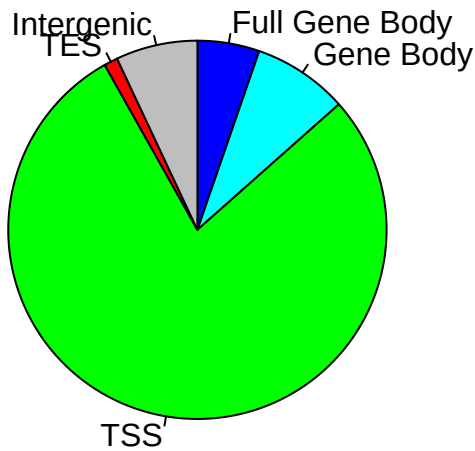
  inside.gene.body <- subset(ostta.genes, seqnames == current.chr &
                             start <= current.start & end >= current.end)
```

```

if(nrow(complete.gene.body) > 0)
{
  target.genes[i] <- paste(unlist(complete.gene.body$gene_id),collapse = ",")
  peak.annotation[i] <- "gene.body"
} else if((nrow(overlap.tss.pos.strand) > 0) | (nrow(overlap.tss.neg.strand) > 0))
{
  target.genes[i] <- paste(c(unlist(overlap.tss.pos.strand$gene_id),
                           unlist(overlap.tss.neg.strand$gene_id)),collapse = ",")
  peak.annotation[i] <- "TSS"
} else if(nrow(overlap.tes.pos.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tes.pos.strand$gene_id),collapse = ",")
  peak.annotation[i] <- "TES"
} else if(nrow(overlap.tes.neg.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tes.neg.strand$gene_id),collapse = ",")
  peak.annotation[i] <- "TES"
} else if(nrow(inside.gene.body) > 0)
{
  target.genes[i] <- paste(unlist(inside.gene.body$gene_id),collapse = ",")
  peak.annotation[i] <- "internal.gene.body"
} else
{
  peak.annotation[i] <- "intergenic"
}
}

freq.peak.annotation <- table(peak.annotation)
#jpeg(filename = "images/h3k4me3_pie_chart_ll_20.jpg",res = 300,width = 900,height = 900)
#par(mar=c(0,0,0,0))
pie(freq.peak.annotation[c("gene.body",
                          "internal.gene.body",
                          "TSS",
                          "TES",
                          "intergenic")],
    labels = c("Full Gene Body",
               "Gene Body",
               "TSS",
               "TES",
               "Intergenic"),
    clockwise = T, border = "black", col=c("blue","cyan","green","red","grey"))

```



```
#dev.off()

percentage.freq.peak.annotation <- 100*freq.peak.annotation/sum(freq.peak.annotation)

percentage.freq.peak.annotation

## peak.annotation
##      gene.body      intergenic internal.gene.body      TES
##      5.304010      6.985770      8.150065      1.164295
##      TSS
##      78.395860

k4.ll.20.peak.annotation <- cbind(k4.ll.20[,1:3],data.frame(peak.annotation,target.genes))

head(k4.ll.20.peak.annotation)

##   chr  start   end peak.annotation      target.genes
## 1   1  12890  13380          TSS      ostta01g00090
## 2   1  34365  35163          TSS ostta01g00220,ostta01g00210
## 3   1  73034  73837          TSS      ostta01g00485
## 4   1 129146 129876          TSS      ostta01g00830
## 5   1 134797 135611          TSS      ostta01g00890
## 6   1 154932 155506          TSS      ostta01g01020

write.table(x = k4.ll.20.peak.annotation,file = "tables/H3K4me3_LL_20C_peak_annotation.tsv",quote = F,s

all.k4.genes <- (unlist(sapply(X =k4.ll.20.peak.annotation$target.genes,
                             FUN = function(x) {strsplit(x,",")[[1]]})))

names(all.k4.genes) <- NULL
length(all.k4.genes)

## [1] 832

length(all.k4.genes)/7668

## [1] 0.1085029
```

We found 832 genes associated to H3K4me3 marks corresponding to 10.9% of the entire *Ostreococcus* genome. H3K4me3 is mostly located occupying the TSS (78.4%) with marginal percentages for internal regions of gene body (8.2%), intergenic regions (7%), entire gene bodies (5.3%) and TES (1.2%).

Next we use metageneplots to determine the shape of the H3K27me3 over its occupied genes. We consider three different sets of marked genes, genes with their entire gene body marked, genes with H3K27 inside their

gene body with no overlap in the TSS or TES and genes with H3K27me3 overlapping their TSS or TES

Next we generate the metagenepLOTS for genes where the H3K27me3 occupies their entire gene body including TSS and TES.

```
library(rtracklayer)
## Genes with their entire gene body marked
k4.gene.body <- (unlist(sapply(X = subset(k4.ll.20.peak.annotation,
                                         peak.annotation == "gene.body")["target.genes"],
                                         FUN = function(x) {strsplit(x, ",")[[1]]})))

names(k4.gene.body) <- NULL
length(k4.gene.body)

## [1] 44

## Genes with H3K27 inside their gene body with no overlap
## in the TSS or TES.
k4.internal.gene.body <- (unlist(sapply(X = subset(k4.ll.20.peak.annotation,
                                                  peak.annotation == "internal.gene.body")["target.genes"],
                                                  FUN = function(x) {strsplit(x, ",")[[1]]})))

names(k4.internal.gene.body) <- NULL
length(k4.internal.gene.body)

## [1] 63

## Genes with H3K27me3 overlapping their TSS
k4.tss <- (unlist(sapply(X = subset(k4.ll.20.peak.annotation,
                                    peak.annotation == "TSS")["target.genes"],
                                    FUN = function(x) {strsplit(x, ",")[[1]]})))

names(k4.tss) <- NULL
length(k4.tss)

## [1] 716

## Genes with H3K27me3 overlapping their TES
k4.tes <- (unlist(sapply(X = subset(k4.ll.20.peak.annotation,
                                    peak.annotation == "TES")["target.genes"],
                                    FUN = function(x) {strsplit(x, ",")[[1]]})))

names(k4.tes) <- NULL
length(k4.tes)

## [1] 9

metagenepLOT.data.k4.gene.body.1 <-
  metagenepLOT(gene.names = k4.gene.body,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k4me3/temp20/h3k4me3_20_chip_1.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k4.gene.body.1 <-
  apply(X = metagenepLOT.data.k4.gene.body.1,
        MARGIN = 2,
        FUN = median, na.rm = T)

metagenepLOT.data.k4.gene.body.2 <-
  metagenepLOT(gene.names = k4.gene.body,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k4me3/temp20/h3k4me3_20_chip_2.bw",
```

```

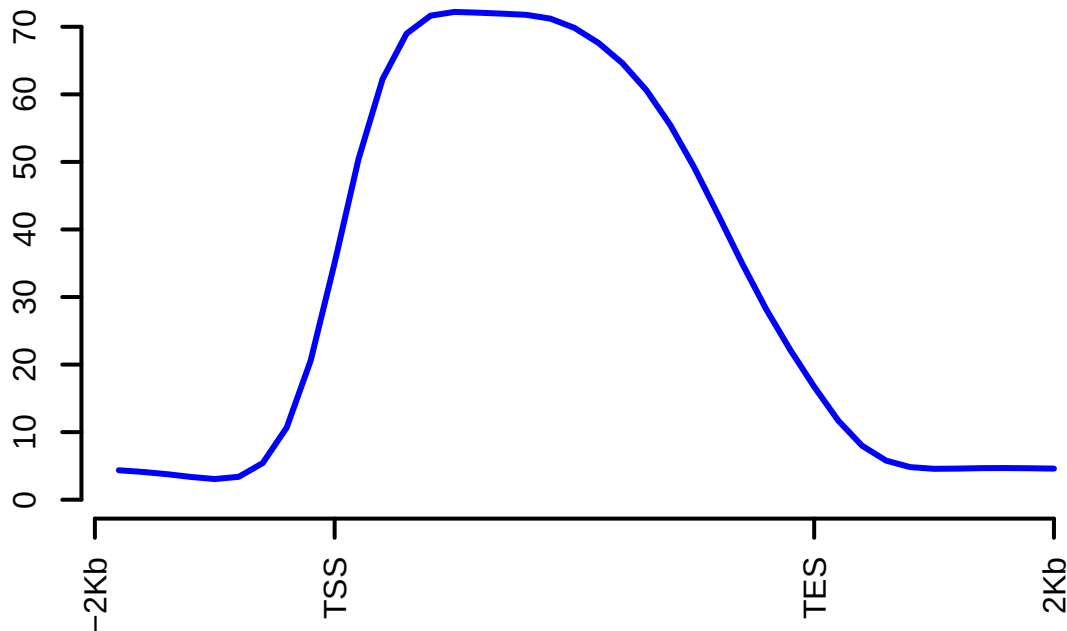
        ntile.txs = 10,
        ntile.gene.body = 20)
median.metagenepLOT.data.k4.gene.body.2 <-
  apply(X = metagenepLOT.data.k4.gene.body.2,
        MARGIN = 2,
        FUN = median,na.rm = T)

median.metagenepLOT.data.k4.gene.body.3 <-
  metagenepLOT(gene.names = k4.gene.body,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k4me3/temp20/h3k4me3_20_chip_3.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k4.gene.body.3 <-
  apply(X = metagenepLOT.data.k4.gene.body.3,
        MARGIN = 2,
        FUN = median,na.rm = T)

median.metagenepLOT.data.k4.gene.body <- (median.metagenepLOT.data.k4.gene.body.1 +
                                           median.metagenepLOT.data.k4.gene.body.2 +
                                           median.metagenepLOT.data.k4.gene.body.3)/3

plot(smooth.spline(x=1:40,y=median.metagenepLOT.data.k4.gene.body,spar = 0.5),
     type="l",col="blue",lwd=3,ylim=c(0,70),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



Next we generate the metagenepLOTS for genes where the H3K4me3 occupies part of the gene body without overlapping with the TSS and TES.

```

metagenepLOT.data.k4.internal.gene.body.1 <-
  metagenepLOT(gene.names = k4.internal.gene.body,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k4me3/temp20/h3k4me3_20_chip_1.bw",

```

```

        ntile.txs = 10,
        ntile.gene.body = 20)
median.metagenepLOT.data.k4.internal.gene.body.1 <-
  apply(X = metagenepLOT.data.k4.internal.gene.body.1,
        MARGIN = 2,
        FUN = median,na.rm = T)

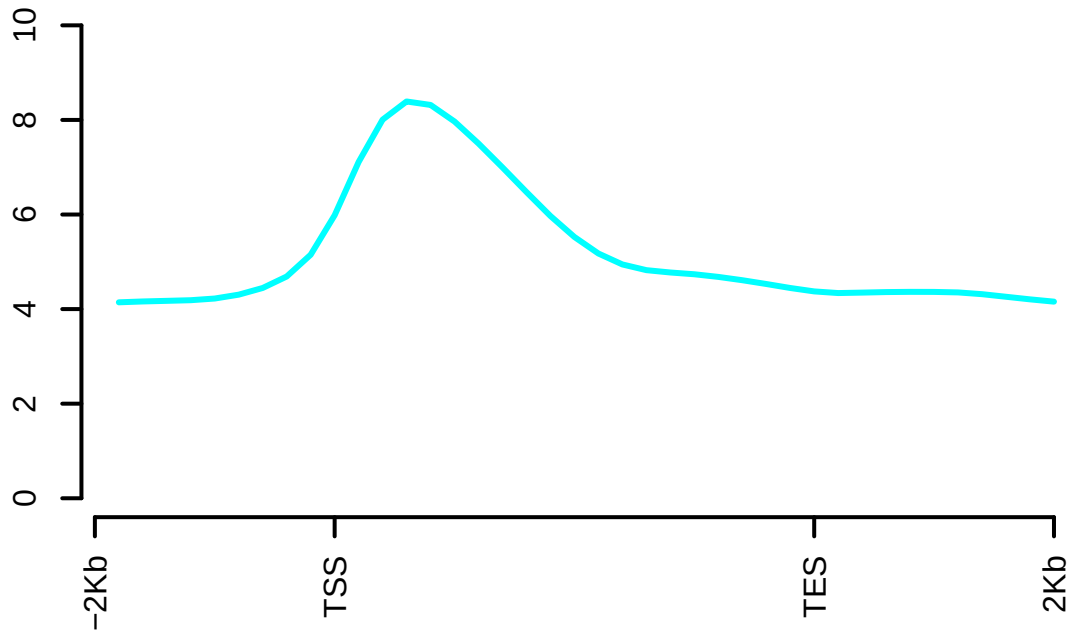
metagenepLOT.data.k4.internal.gene.body.2 <-
  metagenepLOT(gene.names = k4.internal.gene.body,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k4me3/temp20/h3k4me3_20_chip_2.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k4.internal.gene.body.2 <-
  apply(X = metagenepLOT.data.k4.internal.gene.body.2,
        MARGIN = 2,
        FUN = median,na.rm = T)

metagenepLOT.data.k4.internal.gene.body.3 <-
  metagenepLOT(gene.names = k4.internal.gene.body,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k4me3/temp20/h3k4me3_20_chip_3.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k4.internal.gene.body.3 <-
  apply(X = metagenepLOT.data.k4.internal.gene.body.3,
        MARGIN = 2,
        FUN = median,na.rm = T)

median.metagenepLOT.data.k4.internal.gene.body <- (median.metagenepLOT.data.k4.internal.gene.body.1 +
                                                    median.metagenepLOT.data.k4.internal.gene.body.2 +
                                                    median.metagenepLOT.data.k4.internal.gene.body.3)/3

plot(smooth.spline(x=1:40,y=median.metagenepLOT.data.k4.internal.gene.body,spar = 0.5),
     type="l",col="cyan",lwd=3,ylim=c(0,10),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



Next we generate the metagenepLOTS for genes where the H3K4me3 occupies only the TSS and the beginning of the gene.

```
metagenepLOT.data.k4.tss.1 <-
  metagenepLOT(gene.names = k4.tss,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k4me3/temp20/h3k4me3_20_chip_1.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k4.tss.1 <-
  apply(X = metagenepLOT.data.k4.tss.1,
        MARGIN = 2,
        FUN = median, na.rm = T)

metagenepLOT.data.k4.tss.2 <-
  metagenepLOT(gene.names = k4.tss,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k4me3/temp20/h3k4me3_20_chip_2.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k4.tss.2 <-
  apply(X = metagenepLOT.data.k4.tss.2,
        MARGIN = 2,
        FUN = median, na.rm = T)

metagenepLOT.data.k4.tss.3 <-
  metagenepLOT(gene.names = k4.tss,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k4me3/temp20/h3k4me3_20_chip_3.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k4.tss.3 <-
```

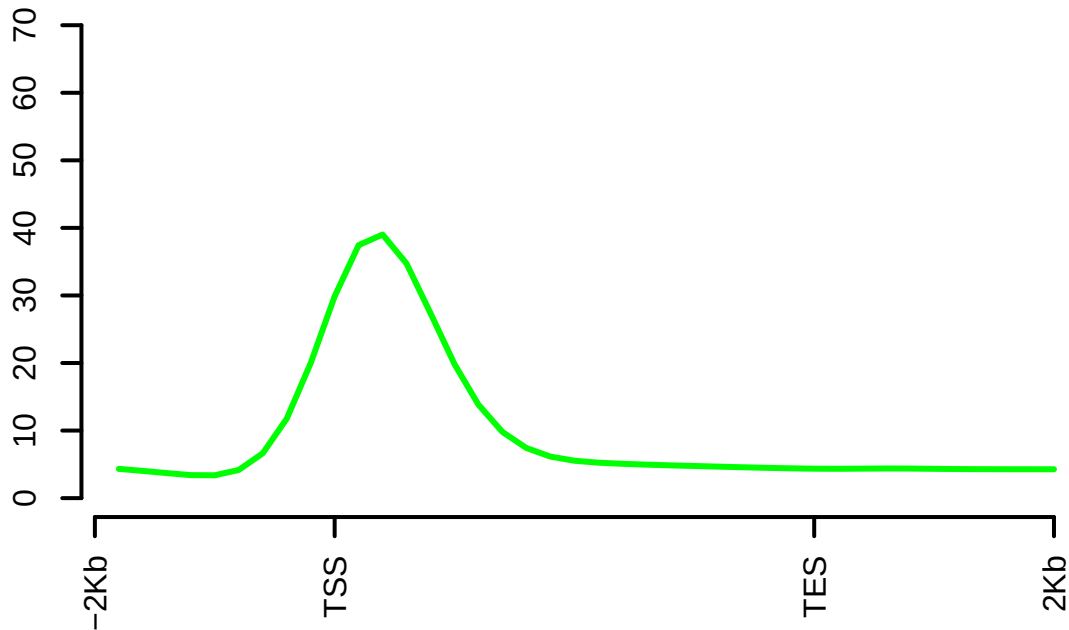
```

apply(X = metageneplot.data.k4.tss.3,
      MARGIN = 2,
      FUN = median, na.rm = T)

median.metageneplot.data.k4.tss <-
  (median.metageneplot.data.k4.tss.1 +
   median.metageneplot.data.k4.tss.2 +
   median.metageneplot.data.k4.tss.3)/3

plot(smooth.spline(x=1:40,y=median.metageneplot.data.k4.tss,spar = 0.5),
     type="l",col="green",lwd=3,ylim=c(0,70),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



Next we generate the metageneplots for genes where the H3K4me3 occupies only the TES and the end of the gene.

```

metageneplot.data.k4.tes.1 <-
  metageneplot(gene.names = k4.tes,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k4me3/temp20/h3k4me3_20_chip_1.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metageneplot.data.k4.tes.1 <-
  apply(X = metageneplot.data.k4.tes.1,
        MARGIN = 2,
        FUN = median, na.rm = T)

metageneplot.data.k4.tes.2 <-
  metageneplot(gene.names = k4.tes,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k4me3/temp20/h3k4me3_20_chip_2.bw",

```

```

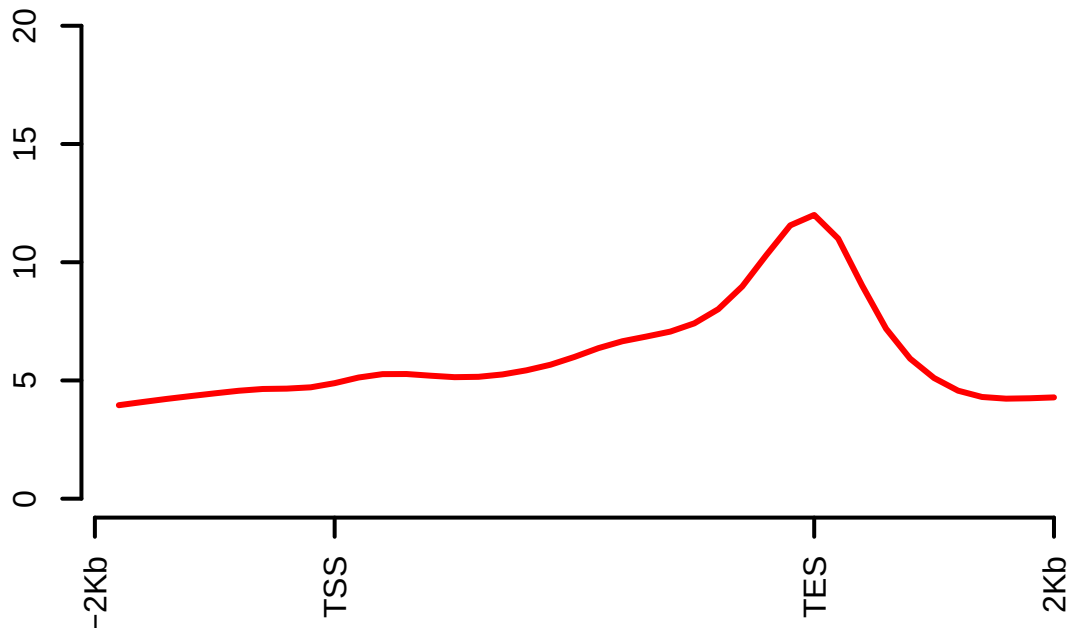
        ntile.txs = 10,
        ntile.gene.body = 20)
median.metagenepLOT.data.k4.tes.2 <-
  apply(X = metagenepLOT.data.k4.tes.2,
        MARGIN = 2,
        FUN = median,na.rm = T)

metagenepLOT.data.k4.tes.3 <-
  metagenepLOT(gene.names = k4.tes,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k4me3/temp20/h3k4me3_20_chip_3.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k4.tes.3 <-
  apply(X = metagenepLOT.data.k4.tes.3,
        MARGIN = 2,
        FUN = median,na.rm = T)

median.metagenepLOT.data.k4.tes <-
  (median.metagenepLOT.data.k4.tes.1 +
   median.metagenepLOT.data.k4.tes.2 +
   median.metagenepLOT.data.k4.tes.3)/3

plot(smooth.spline(x=1:40,y=median.metagenepLOT.data.k4.tes,spar = 0.5),
     type="l",col="red",lwd=3,ylim=c(0,20),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



In order to com-

pare the different metagenepLOTS are represented in the same plot.

```

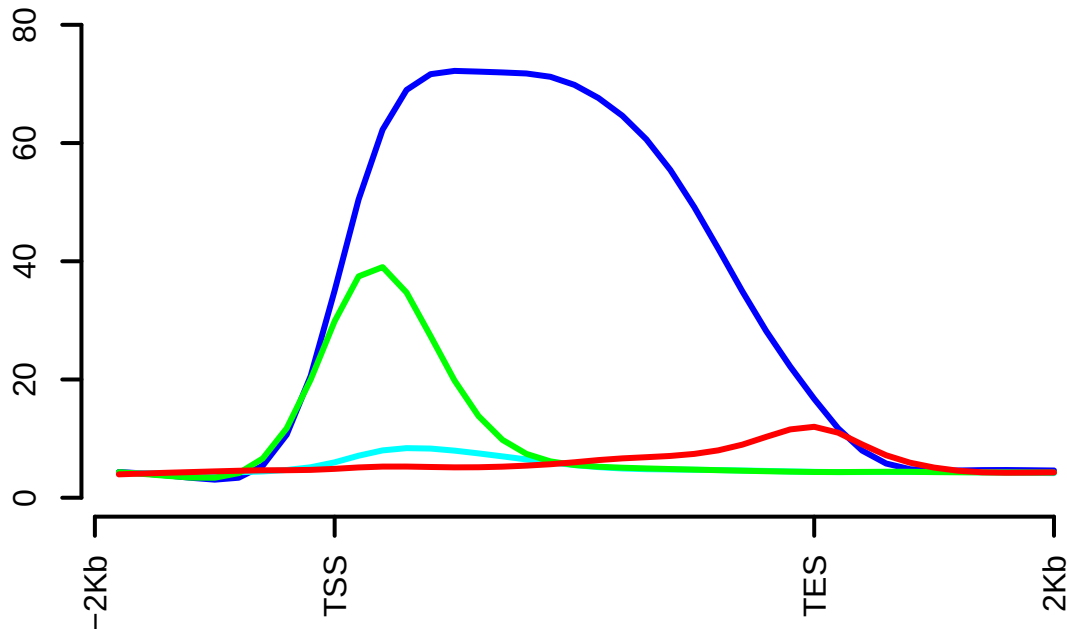
#jpeg(filename = "images/h3k4me3_metagenepLOTS.jpg",res = 250,width = 900,height = 900)
plot(smooth.spline(x=1:40,y=median.metagenepLOT.data.k4.gene.body,spar = 0.5),
     type="l",col="blue",lwd=3,ylim=c(0,80),axes=F,xlab="",ylab="")

```

```

lines(smooth.spline(x=1:40,y=median.metagenesplot.data.k4.internal.gene.body,spar = 0.5),
      type="l",col="cyan",lwd=3)
lines(smooth.spline(x=1:40,y=median.metagenesplot.data.k4.tss,spar = 0.5),
      type="l",col="green",lwd=3)
lines(smooth.spline(x=1:40,y=median.metagenesplot.data.k4.tes,spar = 0.5),
      type="l",col="red",lwd=3)
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



```
#dev.off()
```

It can be observed that the level of H3K4me3 mark varies depending on the part of gene that it occupies. Next we check if these differences are significant.

```

max.metagenesplot.data.k4.gene.body.1 <-
  apply(X = metagenesplot.data.k4.gene.body.1,
        MARGIN = 1,
        FUN = max,na.rm = T)
max.metagenesplot.data.k4.gene.body.2 <-
  apply(X = metagenesplot.data.k4.gene.body.2,
        MARGIN = 1,
        FUN = max,na.rm = T)
max.metagenesplot.data.k4.gene.body.3 <-
  apply(X = metagenesplot.data.k4.gene.body.3,
        MARGIN = 1,
        FUN = max,na.rm = T)

max.metagenesplot.data.k4.gene.body <- (max.metagenesplot.data.k4.gene.body.1 +
                                         max.metagenesplot.data.k4.gene.body.2 +
                                         max.metagenesplot.data.k4.gene.body.3)/3

max.metagenesplot.data.k4.internal.gene.body.1 <-
  apply(X = metagenesplot.data.k4.internal.gene.body.1,
        MARGIN = 1,

```

```

    FUN = max, na.rm = T)
max.metagenepplot.data.k4.internal.gene.body.2 <-
  apply(X = metagenepplot.data.k4.internal.gene.body.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k4.internal.gene.body.3 <-
  apply(X = metagenepplot.data.k4.internal.gene.body.3,
        MARGIN = 1,
        FUN = max, na.rm = T)

max.metagenepplot.data.k4.internal.gene.body <- (max.metagenepplot.data.k4.internal.gene.body.1 +
  max.metagenepplot.data.k4.internal.gene.body.2 +
  max.metagenepplot.data.k4.internal.gene.body.3)/3

max.metagenepplot.data.k4.tss.1 <-
  apply(X = metagenepplot.data.k4.tss.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k4.tss.2 <-
  apply(X = metagenepplot.data.k4.tss.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k4.tss.3 <-
  apply(X = metagenepplot.data.k4.tss.3,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k4.tss <- (max.metagenepplot.data.k4.tss.1 +
  max.metagenepplot.data.k4.tss.2 +
  max.metagenepplot.data.k4.tss.3)/3

max.metagenepplot.data.k4.tes.1 <-
  apply(X = metagenepplot.data.k4.tes.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k4.tes.2 <-
  apply(X = metagenepplot.data.k4.tes.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k4.tes.3 <-
  apply(X = metagenepplot.data.k4.tes.3,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k4.tes <- (max.metagenepplot.data.k4.tes.1 +
  max.metagenepplot.data.k4.tes.2 +
  max.metagenepplot.data.k4.tes.3)/3

shapiro.test(max.metagenepplot.data.k4.gene.body)

##
##  Shapiro-Wilk normality test
##
## data:  max.metagenepplot.data.k4.gene.body
## W = 0.66106, p-value = 8.089e-09

```

```

shapiro.test(max.metagenesplot.data.k4.internal.gene.body)

##
##  Shapiro-Wilk normality test
##
## data:  max.metagenesplot.data.k4.internal.gene.body
## W = 0.54737, p-value = 1.21e-12
shapiro.test(max.metagenesplot.data.k4.tss)

##
##  Shapiro-Wilk normality test
##
## data:  max.metagenesplot.data.k4.tss
## W = 0.77699, p-value < 2.2e-16
shapiro.test(max.metagenesplot.data.k4.tes)

##
##  Shapiro-Wilk normality test
##
## data:  max.metagenesplot.data.k4.tes
## W = 0.78482, p-value = 0.01368
wilcox.test(x = max.metagenesplot.data.k4.gene.body,
            y = max.metagenesplot.data.k4.internal.gene.body,
            alternative = "greater")$p.value

## [1] 4.21873e-11
wilcox.test(x = max.metagenesplot.data.k4.gene.body,
            y = max.metagenesplot.data.k4.tss,
            alternative = "greater")

##
##  Wilcoxon rank sum test with continuity correction
##
## data:  max.metagenesplot.data.k4.gene.body and max.metagenesplot.data.k4.tss
## W = 22634, p-value = 5.621e-07
## alternative hypothesis: true location shift is greater than 0
wilcox.test(x = max.metagenesplot.data.k4.gene.body,
            y = max.metagenesplot.data.k4.tes,
            alternative = "greater")

##
##  Wilcoxon rank sum exact test
##
## data:  max.metagenesplot.data.k4.gene.body and max.metagenesplot.data.k4.tes
## W = 345, p-value = 9.7e-05
## alternative hypothesis: true location shift is greater than 0
wilcox.test(x = max.metagenesplot.data.k4.tss,
            y = max.metagenesplot.data.k4.tes)

##
##  Wilcoxon rank sum test with continuity correction
##

```

```
## data: max.metagenesplot.data.k4.tss and max.metagenesplot.data.k4.tes
## W = 4819, p-value = 0.01056
## alternative hypothesis: true location shift is not equal to 0
```

Indeed the differences between all metagenesplots are highly significant.

Now we analyse the gene expression level of H3K4me3 marked genes.

```
non.k4.genes <- setdiff(names(gene.expression.20),c(k4.gene.body,k4.tss,k4.tes,k4.internal.gene.body))
```

```
summary(gene.expression.20[k4.gene.body])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
##    1.184    71.913    289.375    853.606    880.687   13604.318
```

```
summary(gene.expression.20[k4.internal.gene.body])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
##    0.099    35.021    63.333    413.565    196.045   10865.045
```

```
summary(gene.expression.20[k4.tss])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
##    0.004    38.296   127.864   394.627   463.496   9906.184
```

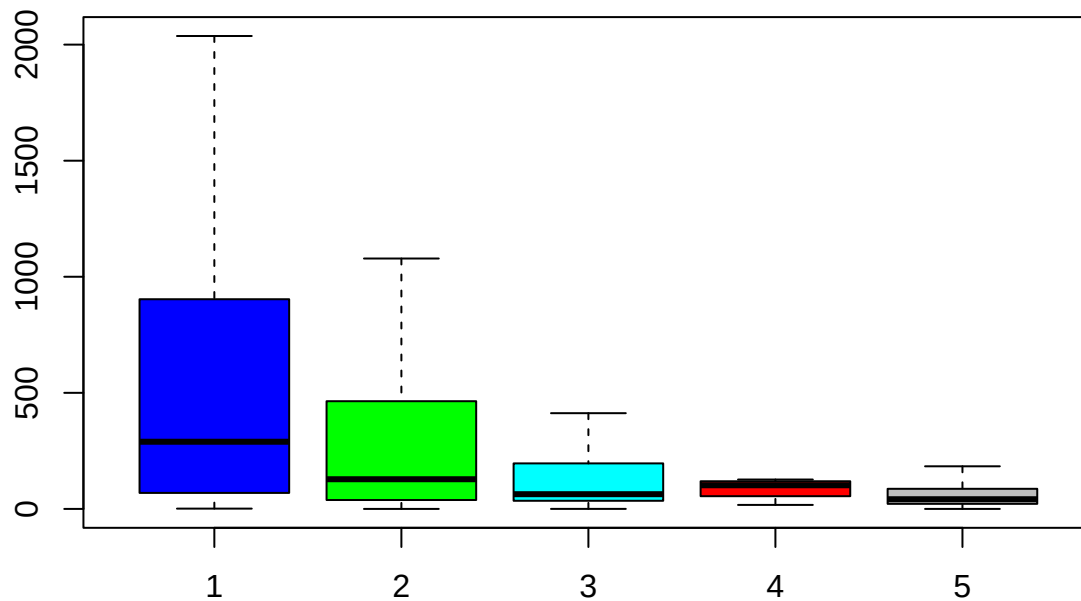
```
summary(gene.expression.20[k4.tes])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
##   17.14    54.79   101.82   129.93   119.14    541.46
```

```
summary(gene.expression.20[non.k4.genes])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
##    0.004    21.744    41.692    83.634    86.522   6961.033
```

```
#jpeg(filename = "images/h3k24me3_expression_barplot_different_genes.jpg",res = 250,width = 600,height = 400)
boxplot(gene.expression.20[k4.gene.body],
        gene.expression.20[k4.tss],
        gene.expression.20[k4.internal.gene.body],
        gene.expression.20[k4.tes],
        gene.expression.20[non.k4.genes],
        col=c("blue","green","cyan","red","grey"),
        outline=F)
```



```
#dev.off()
```

```
wilcox.test(x = gene.expression.20[k4.gene.body],
            y = gene.expression.20[k4.tss],
            alternative = "greater")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k4.gene.body] and gene.expression.20[k4.tss]
## W = 19186, p-value = 0.007567
## alternative hypothesis: true location shift is greater than 0
```

```
wilcox.test(x = gene.expression.20[k4.tss],
            y = gene.expression.20[k4.internal.gene.body],
            alternative = "greater")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k4.tss] and gene.expression.20[k4.internal.gene.body]
## W = 26406, p-value = 0.01225
## alternative hypothesis: true location shift is greater than 0
```

```
wilcox.test(x = gene.expression.20[k4.internal.gene.body],
            y = gene.expression.20[k4.tes],
            alternative = "greater")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k4.internal.gene.body] and gene.expression.20[k4.tes]
## W = 272, p-value = 0.5809
## alternative hypothesis: true location shift is greater than 0
```

```
wilcox.test(x = gene.expression.20[k4.tes],
            y = gene.expression.20[non.k4.genes],
```

```

alternative = "greater")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k4.tes] and gene.expression.20[non.k4.genes]
## W = 42047, p-value = 0.02895
## alternative hypothesis: true location shift is greater than 0
wilcox.test(x = gene.expression.20[k4.gene.body],
            y = gene.expression.20[non.k4.genes],
            alternative = "greater")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k4.gene.body] and gene.expression.20[non.k4.genes]
## W = 236961, p-value = 2.505e-11
## alternative hypothesis: true location shift is greater than 0
wilcox.test(x = gene.expression.20[k4.internal.gene.body],
            y = gene.expression.20[non.k4.genes],
            alternative = "greater")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k4.internal.gene.body] and gene.expression.20[non.k4.genes]
## W = 274724, p-value = 8.714e-05
## alternative hypothesis: true location shift is greater than 0
wilcox.test(x = gene.expression.20[k4.tss],
            y = gene.expression.20[non.k4.genes],
            alternative = "greater")$p.value

## [1] 1.702554e-70
wilcox.test(x = gene.expression.20[k4.tes],
            y = gene.expression.20[non.k4.genes],
            alternative = "greater")$p.value

## [1] 0.02895157

```

We can see all H3K4me3 genes present levels of expression significantly higher than non-marked genes. Specifically, genes whose entire gene body or TSS is occupied by H3K4me3 are much higher expressed. Whereas, genes where only the TES or gene body is occupied by H3K4me3 are expressed at a significantly higher level than non-marked genes but with much lower significance level.

In what follows we perform a functional analysis over the H3K4me3 marked genes. First we study the enrichment in GO terms.

```

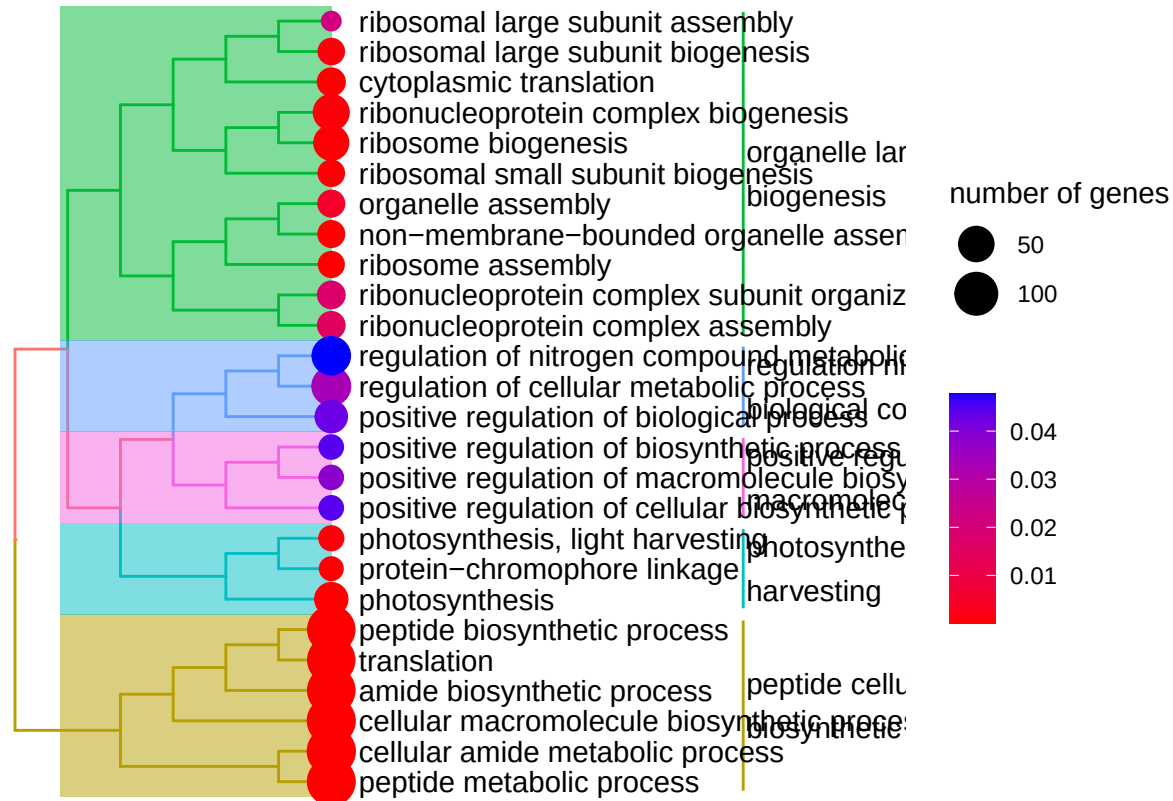
library(clusterProfiler)
library(enrichplot)
library(org.Obtativ5.eg.db)
k4.enrichment <- enrichGO(gene = all.k4.genes, pvalueCutoff = 0.05, qvalueCutoff = 0.05,
                          OrgDb = org.Obtativ5.eg.db,
                          ont = "BP",
                          keyType = "GID")

```

```
k4.enrichment.df <- as.data.frame(k4.enrichment)

write.table(x = k4.enrichment.df, file = "go_enrichment_k4.tsv", quote = F, sep = "\t", row.names = F)

treeplot(x = pairwise_termsim(k4.enrichment))
```



```
library(treemap)

revigo.names <- c("term_ID", "description", "frequency", "value", "uniqueness", "dispensability", "representativeness")
revigo.data <- rbind(c("GO:0006412", "translation", 4.38869169324396, 41.635159951405136, 0.687584154330528, 0.012821578346646382, 3.275341540815408),
  c("GO:0002181", "cytoplasmic translation", 0.3974763229665376, 8.595948611417302, 0.7511417302, 0.012821578346646382, 3.275341540815408),
  c("GO:0009059", "macromolecule biosynthetic process", 16.15219652221219, 32.13388504719652, 0.012821578346646382, 3.275341540815408),
  c("GO:0043603", "amide metabolic process", 6.707376287050344, 36.45011550587449, 0.910587449, 0.012821578346646382, 3.275341540815408),
  c("GO:0009765", "photosynthesis, light harvesting", 0.02241434707504848, 4.839062538504848, 0.02241434707504848, 0.012821578346646382, 3.275341540815408),
  c("GO:0015979", "photosynthesis", 0.228607115192195, 7.075542911759376, 0.941812778414181, 0.012821578346646382, 3.275341540815408),
  c("GO:0048518", "positive regulation of biological process", 4.759580864033397, 2.9264033397, 0.012821578346646382, 3.275341540815408),
  c("GO:0010557", "positive regulation of macromolecule biosynthetic process", 2.0931810557, 2.0931810557, 0.012821578346646382, 3.275341540815408),
  c("GO:0031323", "regulation of cellular metabolic process", 14.819661863282441, 3.0711863282441, 0.012821578346646382, 3.275341540815408),
  c("GO:0140694", "non-membrane-bounded organelle assembly", 0.7232459602662196, 6.42502662196, 0.012821578346646382, 3.275341540815408),
  c("GO:0000027", "ribosomal large subunit assembly", 0.012821578346646382, 3.275341540815408, 0.012821578346646382, 3.275341540815408),
  c("GO:0022613", "ribonucleoprotein complex biogenesis", 2.5375929564430133, 4.856610133, 0.012821578346646382, 3.275341540815408),
  c("GO:0042254", "ribosome biogenesis", 2.136224808753416, 5.364416283328431, 0.566141516, 0.012821578346646382, 3.275341540815408),
  c("GO:0070925", "organelle assembly", 1.3561320812847646, 3.9914667506842227, 0.621342227, 0.012821578346646382, 3.275341540815408),
  c("GO:0071826", "protein-RNA complex organization", 0.6518695497816881, 3.39757827104, 0.012821578346646382, 3.275341540815408))

stuff <- data.frame(revigo.data);
names(stuff) <- revigo.names;
```

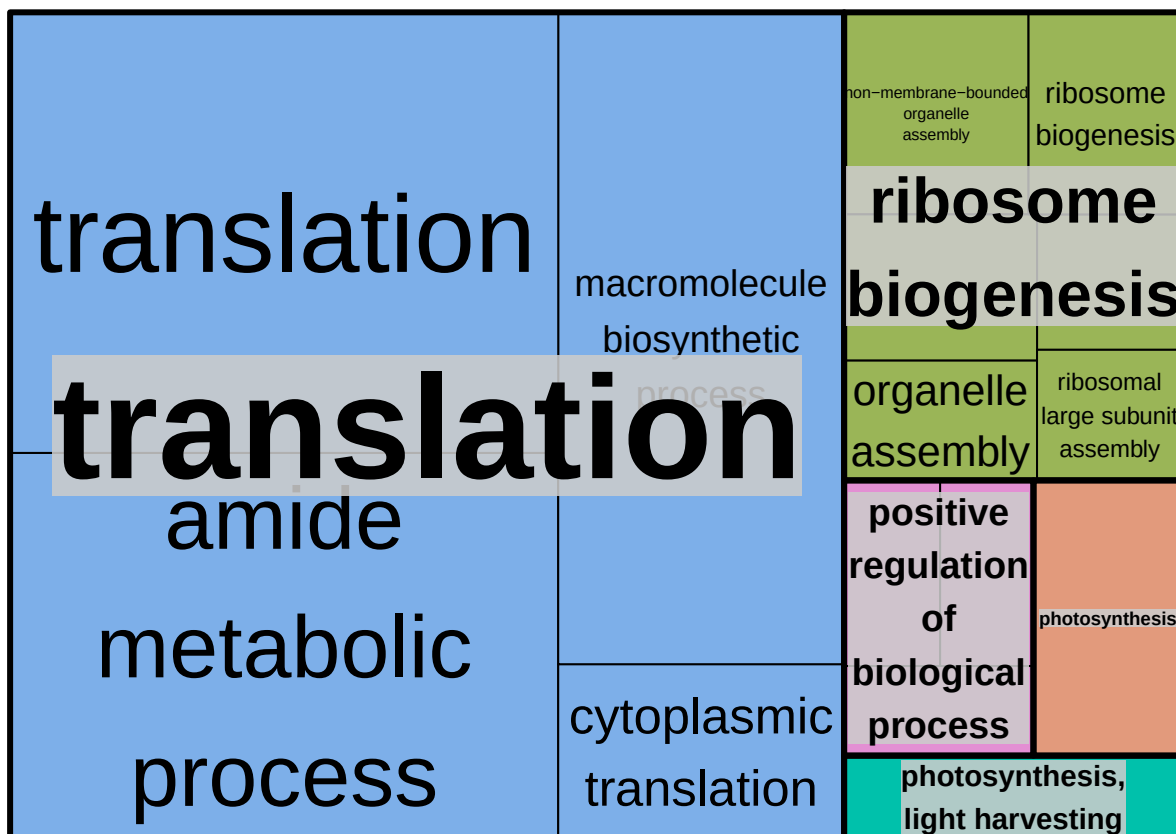
```

stuff$value <- as.numeric( as.character(stuff$value) );
stuff$frequency <- as.numeric( as.character(stuff$frequency) );
stuff$uniqueness <- as.numeric( as.character(stuff$uniqueness) );
stuff$dispensability <- as.numeric( as.character(stuff$dispensability) );

#jpeg(filename = "images/h3k4me3_revigo_treemap.jpg",res = 300,width = 1400,height = 900)

treemap(
  stuff,
  index = c("representative","description"),
  vSize = "value",
  type = "categorical",
  vColor = "representative",
  title = "",
  inflate.labels = TRUE,
  lowerbound.cex.labels = 0,
  position.legend = "none"
)

```



```
#dev.off()
```

H3K27me3 and H3K4me3 common marked genes

The set of H3K27me3 and H3K4me3 genes only overlapped marginally.

```
overlap.k4.k27 <- read.table(file="chipseq_data/bed_data/h3k27me3_h3k4me3_11_20.bed",header=F,sep = "\t")
```

```

max(overlap.k4.k27$V3 - overlap.k4.k27$V2)

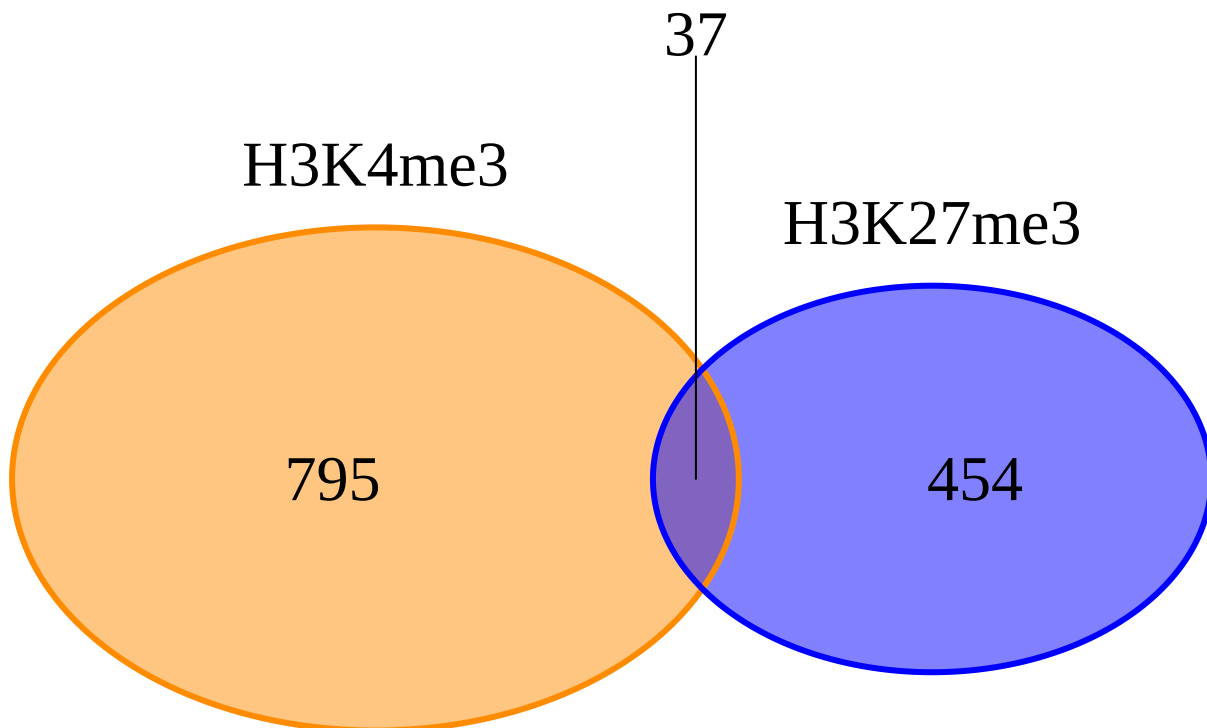
## [1] 411
summary(overlap.k4.k27$V3 - overlap.k4.k27$V2)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      22.0   89.0   199.0   191.2   286.5   411.0
nrow(overlap.k4.k27)

## [1] 19
library(VennDiagram)

## Loading required package: grid
## Loading required package: futile.logger
#jpeg(filename = "images/h3k27me3_h3k4me3_venndiagram.jpg",res = 250,width = 1500,height = 1500)
grid.newpage()
draw.pairwise.venn(area1 = length(all.k27.genes),cat.pos = c(180,180),cat.dist = 0.05,
  area2 = length(all.k4.genes),
  cross.area = length(intersect(all.k27.genes,all.k4.genes)),
  lwd = 3,category = c("H3K27me3","H3K4me3"),euler.d = T,
  col = c("blue","darkorange"),
  fill = c(" blue","darkorange"),alpha = 0.5,
  cex = 2,
  cat.cex = 2)

```



```

## (polygon[GRID.polygon.608], polygon[GRID.polygon.609], polygon[GRID.polygon.610], polygon[GRID.polygon.611])
#dev.off()

```

Only 19 genomic regions were found simultaneously marked by H3K27me3 and H3K4me3. The length of the

overlap was never longer than 400bp and on average was less than 200bp. Taking into account that the length of DNA wrapped around a single nucleosome is approximately 150bp, no more than one or two nucleosomes appeared marked simultaneously by H3K27me3 and H3K4me3.

Only 37 genes were identified as bivalent simultaneously marked with H3K27me3 and H3K4me3. Next we explore the positioning of these two marks in these genes and their level of expression.

```
k4.k27.genes <- intersect(all.k4.genes, all.k27.genes)
k4.k27.gene.annotation <- subset(11.20.peak.annotation, target.genes %in% k4.k27.genes)
100*table(k4.k27.gene.annotation[!duplicated(k4.k27.gene.annotation$target.genes),][["peak.annotation"]])
```

```
##
##          gene.body internal.gene.body          TES          TSS
##          9.756098         58.536585        14.634146         4.878049
```

In most bivalent genes H3K27me3 is located in inner regions of gene bodies (58.5%). In 14.6% of the bivalent genes H3K27me3 is located only at the TES. Whereas in only 9.8% and 4.9% H3K27me3 occupies the entire gene body or only the TSS, respectively.